Space Interlayer Conjecture / Phase Manifestation Mechanics (SIC/PMM)

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Modern physics faces a structural limitation in its pursuit of unification, not due to a lack of models, but due to a set of deeply embedded and mutually coupled foundational assumptions. These include the choice of numerical base systems, measurement units, the linear treatment of time, the assumption of spacetime continuity, and the reliance on absolute symmetry.

This work demonstrates that these assumptions are not independent, but form a logically interlocked system in which modifying any single component inevitably produces contradictions in the others.By systematically removing these constraints, we show that a unified phase-based framework—Phase Manifestation Mechanics (PMM)—is not introduced, but **necessarily derived**. This framework is governed by a single nonlinear constant, $\beta \approx 0.0865$, from which physical structure emerges.

Within this formulation, physical quantities are reinterpreted as manifestations of underlying phase structures, rather than independent entities.Under this reconstruction, interference, quantization, symmetry breaking, and temporal dynamics emerge as geometric consequences of phase evolution and projection, rather than fundamental postulates.This suggests that physical laws are not primary descriptors of reality, but stable manifestations of phase structures under constrained geometric conditions.

Keywords: SIC / PMM Framework, Autonomous Cosmological Generative Algorithm, Derivation of Dimensionless Fundamental Constants, Discrete Computational Operators, Geometric Manifestation of Causality, beta-Offset Phase Projection Dynamics, Hubble Constant Tension, Non-Euclidean Quantum Geometry, Numerical Base-System Rectification, Symmetry Breaking and Emergent Temporal Asymmetry.

INTRODUCTION

Modern physics faces a structural limitation in its pursuit of unification, not due to a lack of models, but due to a set of deeply embedded and mutually coupled foundational assumptions. These include the choice of numerical base systems, measurement units, the linear treatment of time, the assumption of spacetime continuity, and the reliance on absolute symmetry.

This work demonstrates that these assumptions are not independent, but form a logically interlocked system in which modifying any single component inevitably produces contradictions in the others. By systematically removing these constraints, we show that a unified phase-based framework—Phase-Manifestation Mechanics (PMM)—is not introduced, but **necessarily derived**. This framework is governed by a single nonlinear constant, $\beta \approx 0.0865$, from which physical structure emerges.

> **This framework defines "reality" as the manifestation of a background phase under finite capacity and resolution constraints.**

This work is built upon the **Spatial Interlayer Equations (SIE)**, aiming to provide a foundational calibration across classical mechanics, general relativity, and quantum mechanics. Unlike conventional approaches in physics—where theoretical models are constructed first and later matched to experimental observations—SIC incorporates **observability** directly into the generative conditions of the framework. By adopting a dimensionless structural basis, it establishes a geometric system that simultaneously includes both a **generative mechanism** and a **manifestation filtering process**.

The objective of this framework is not to replace existing physical theories, but rather to:

> **Define the geometric conditions under which these theories hold, and explain their consistency of manifestation across different scales.**

During the initial derivation, a systematic deviation emerged when the framework was compared against established experimental data. This deviation was not treated as an error, but instead interpreted as an indication of structural elements not yet accounted for in the original formulation. Accordingly, the framework was further refined and extended from SIC into:

> **Phase-Manifestation Mechanics (PMM)**

This refinement introduces **phase manifestation** as the central descriptive principle, enabling improved consistency and explanatory alignment with existing experimental observations. PMM is therefore adopted as the finalized form of the framework.

Finally, it should be noted that throughout the development and organization of this work, large language models (LLMs) were employed as auxiliary tools. Their roles include:

- Rapid cross-domain information comparison
- Semantic alignment and restructuring of concepts
- Acceleration of convergence within dimensionless formulations

LLMs do not contribute to the generation of the theory itself. Their function is limited to:

> **Assisting in the identification of latent structural correspondences within high-dimensional data spaces, and improving the efficiency of the derivation process.**

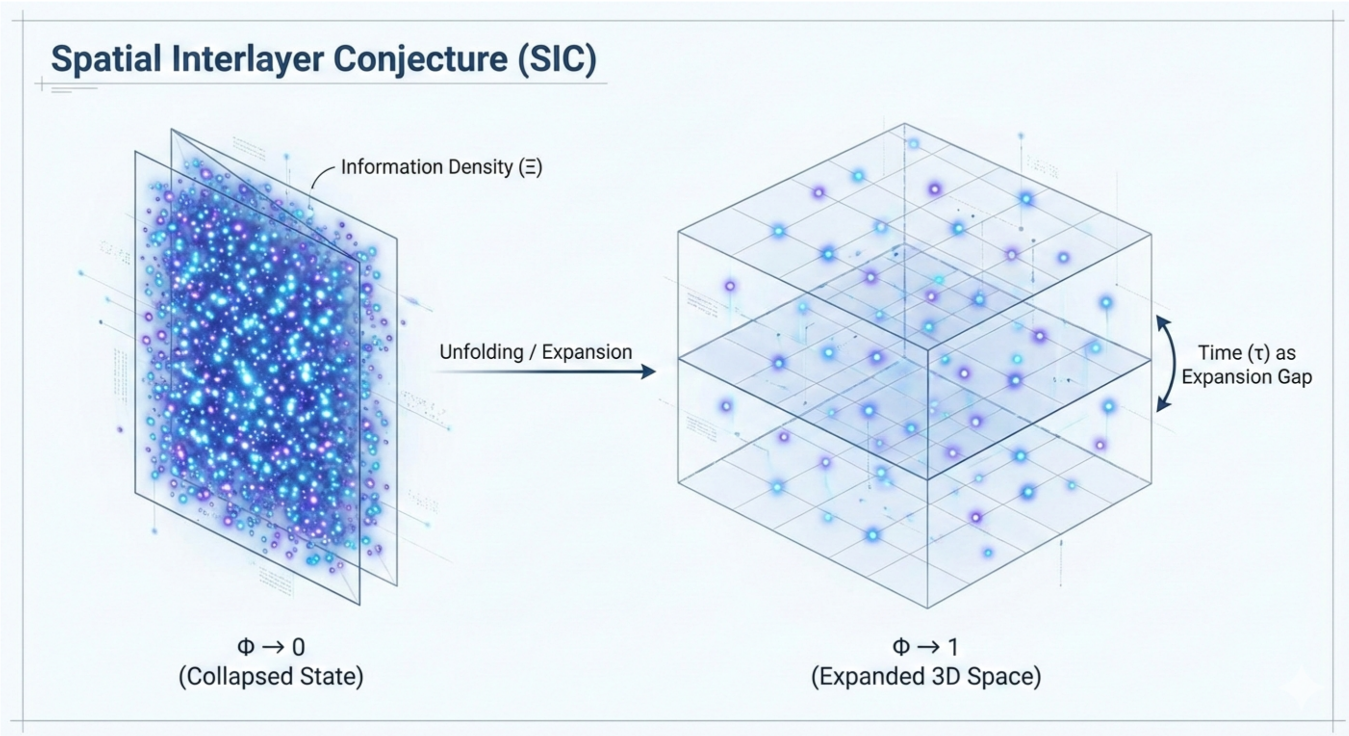

FIG. 1. Projection mechanism of spatial interlacing and information density (Ξ), depicting the geometric state under the condition of $\Phi \to 0$.

CORE PRINCIPLE: INDETERMINACY AND SPONTANEOUS UNFOLDING

To establish a physical framework with intrinsic generative capacity, SIC introduces a fundamental starting point:

In the limit where spatial structure has not yet unfolded ($\Phi \to 0$), the system does not correspond to "zero," but instead enters an **indeterminate and fully overlapped geometric state**.

In this state, information (Ξ) is non-distinguishable, and all possible configurations are geometrically equivalent. This represents a condition of **maximal degrees of freedom combined with complete indistinguishability**, forming a degenerate structure.

However, this degeneracy is not stable. Rather, it is an inherently unsustainable configuration.

$$\Phi \to 0 \Rightarrow \text{Indeterminate / Degenerate State} \tag{1}$$

$$\text{Break of Degeneracy} \Rightarrow \Phi > 0 \Rightarrow \text{Geometric Expansion} \tag{2}$$

Core Mechanism:

Once any infinitesimal asymmetry arises, the system can no longer maintain the fully overlapped indeterminate state. It must resolve into a set of distinguishable geometric configurations.

This transition constitutes the origin of both spatial unfolding and the emergence of time.

> **Thus, the expansion of the universe is not driven by an external force, but is a necessary consequence of the geometric instability of an indeterminate, fully overlapped state.**

In mathematical terms, this corresponds to a limiting case of **breakdown of definability**.

Structures that are traditionally expressed as divergences or undefined quantities are, within the SIC framework, reinterpreted as:

> **signals of geometric phase transition, rather than failures of the theory.**

AXIOMATIC SYSTEM: GEOMETRIC FOUNDATIONS OF SIC

To ensure structural consistency and extensibility, SIC does not begin from existing physical laws. Instead, it is constructed upon a minimal and closed set of geometric axioms.All physical phenomena are treated as manifestations of these axioms under different conditions.

Axiom I: Information Conservation

Information density (Ξ) is the only fundamental conserved quantity in the universe.All physical quantities—such as energy, mass, and fields—are interpreted as different geometric representations of information under varying structural configurations.

Axiom II: Expansion Factor

The physical properties of space are determined by the expansion parameter ($\Phi \in [0, 1]$).

- $\Phi \to 0$ corresponds to a fully overlapped, indeterminate state
- $\Phi \to 1$ corresponds to a stable macroscopic geometric structure

Axiom III: Temporal Cost

Time (τ) is not an independent dimension, but represents the geometric cost required for information to transition from an overlapped state into a distinguishable configuration.

Axiom IV: Discrete Settlement

Physical state changes do not occur through continuous evolution, but through discrete settlement events when geometric conditions are satisfied.Continuity emerges only as a statistical approximation under high-frequency settlement.

Axiom V: Observation Operator

Observation is not an external act, but an intrinsic constraint operator embedded within the geometric structure.It determines which configurations of information are permitted to manifest as observable reality.

Axiom VI: Nonlocal Projection

In the microscopic limit ($\Phi \to 0$), information (Ξ) exists in a fully overlapped and indistinguishable state. At this level, spatial distance (D) and time (τ) are not defined. So-called "nonlocal phenomena" (e.g., quantum entanglement) are therefore not the result of superluminal information transfer, but rather:

> **the simultaneous manifestation of the same underlying information structure at different macroscopic geometric locations.**

Within this framework:

- **No transmission across distance exists**
  (since distance has not yet emerged, $\mathbb{D} = 0$)
- **No temporal ordering exists**
  (since derived time has not yet been incurred, $\tau = 0$)
- **Causality arises as a manifestation constraint**
  The causal sequences observed macroscopically are not due to underlying transmission, but are imposed by the ordering action of the observation operator.

Structural Implication

The apparent correlations between events in the macroscopic world do not originate from the propagation of information, but from:

> **The consistent selection of a shared underlying information structure under different geometric conditions.These axioms do not describe phenomena; they define the conditions under which phenomena can exist.Any physical theory that cannot be reconstructed within this geometric framework does not access the underlying structure.**

CORE GEOMETRIC VARIABLES

SIC rejects the notion of space as a passive background. Instead, it proposes that physical reality emerges as a geometric manifestation of underlying **information**, modulated by the **spatial expansion parameter (Φ)**.

Within this framework, four fundamental geometric variables are defined. These variables form a **closed coupled system**, while maintaining a well-defined directional structure in their generative sequence:

> **Generative Order:**
> $\Phi \ \rightarrow \ \Xi \ \rightarrow \ \mathbb{D} \ \rightarrow \ \tau$

Φ — Spatial Expansion Parameter

$\Phi \in [0, 1]$ is the primary variable governing the physical characteristics of the interlayer structure.

It describes the geometric state of information ranging from complete overlap to full expansion:

- $\Phi = 0$: fully collapsed, singular overlapped state
- $\Phi > 0$: emergence of distinguishable geometric structure

All subsequent physical quantities arise only after deviation from the $\Phi = 0$ state.

Ξ — Information Overlap Density

Ξ characterizes the distribution of information as determined by Φ.

It represents the effective information density carried by a given geometric configuration under specific expansion conditions. Within the SIC framework, Ξ is globally conserved and serves as the primary constraint governing geometric evolution.

$\mathbb{D}$ — Interlayer Distance

$\mathbb{D}$ emerges when information density (Ξ) becomes geometrically separated under non-zero expansion.

It does not constitute a fundamental quantity, but rather arises as a consequence of structural differentiation:

- As $\Phi \rightarrow 0$, $\mathbb{D} \rightarrow 0$
- Spatial separation ceases to exist in the fully overlapped limit

τ — Derivative Time

τ represents the geometric cost associated with transitions between different Φ states.

Time is therefore not a background dimension, but an emergent quantity resulting from structural reconfiguration. Its magnitude is jointly determined by:

- the degree of geometric change ($\Delta\Phi$)
- the extent of information redistribution ($\Delta\Xi$)

Structural Note

Although these variables form a closed coupled system (e.g., τ constrains the rate of change of Φ), their generative logic is not symmetric.

SIC asserts that all observable physical quantities originate from the initial deviation of Φ away from the fully overlapped state, triggering a cascading structural unfolding.

PRIMITIVE CONSTRAINT STRUCTURE (SIE: SPATIAL INTERLAYER EQUATIONS)

These variables are not merely descriptive. Within the SIC framework, Φ, Ξ, $\mathbb{D}$, and τ satisfy a set of **primitive constraint relations** that precede any specific physical phenomena.

These relations:

- do not depend on predefined notions of particles, energy, or fields
- operate directly on geometric and informational structures

The central principle is:

> **All observable physical quantities must emerge as consistent solutions under these constraints, rather than being independently postulated.**

In other words, SIC does not assume existing physical laws. Instead, it restricts the allowable relationships between variables, such that known physical theories arise as effective representations under specific scales and conditions.

Based on this foundation, the formal **SIE equations** will be introduced to quantify the relationships between information density, spatial structure, and temporal cost, serving as the basis for empirical testability.

SIE-1: Fundamental Constraint Between Information Density and Spatial Expansion

$$\Xi(\Phi) = \frac{\Xi_0}{\Phi + \epsilon} \tag{3}$$

Structural Description:
This equation defines the fundamental constraint between **information overlap density** (Ξ) and the **spatial expansion rate** (Φ). As $\Phi \to 0$, the system approaches a highly overlapped state, and the information density diverges. As Φ increases, Ξ decreases monotonically, corresponding to the emergence of geometric separation and distinguishability.

Parameter Definitions:

- Ξ_0: Baseline information scale, representing the characteristic information intensity under maximal overlap conditions.
- ϵ: Regularization parameter, corresponding to the minimum resolvable geometric scale, preventing divergence at $\Phi = 0$.

Physical Implications:

- Establishes a monotonic **expansion–dilution correspondence**, directly coupling spatial structure formation with information distribution.
- As $\Phi \to 0$, the system enters a high-density unstable regime, providing the intrinsic drive for spontaneous geometric expansion.
- This relation is independent of specific physical fields or particle models, and serves as a **primitive geometric constraint** within the SIC framework.

SIE-2: Generation Law of Spatial Scale

$$\mathbb{D}(\Phi) = D_0 \cdot \Phi^{\alpha} \tag{4}$$

Structural Description:
This equation describes how **interlayer distance** ($\mathbb{D}$) emerges as a function of the spatial expansion rate (Φ).
When $\Phi = 0$, space has not unfolded and distance is undefined ($\mathbb{D} \to 0$). As Φ increases, geometric separation forms progressively, and distance emerges as a measurable scale of information separation.
Parameter Definitions:

- D_0: Baseline spatial scale, setting the unit of distance in the macroscopic expansion regime.
- α: Geometric expansion exponent, characterizing the nonlinearity of spatial emergence.

Physical Implications:

- Spatial distance is not fundamental, but a **derived structure** arising from the decoupling of overlapping information states.
- Establishes the generative mechanism: **expansion → distinguishable distance**.
- As $\Phi \to 0$, distance collapses, providing a geometric basis for **nonlocal phenomena**.

SIE-3: Time as the Geometric Cost of Information Transfer

$$\tau = \int \mathcal{C}(\Phi)\, d\mathbb{D} \tag{5}$$

Structural Description:
This equation defines **derivative time** (τ) not as an accumulation of state changes, but as the **geometric delay cost** required for information to propagate across an already expanded spatial structure.
Function Definition:

- $\mathcal{C}(\Phi)$: Geometric cost function, describing the minimum delay required for information to traverse interlayer distance ($\mathbb{D}$) under a given spatial expansion rate.

Physical Implications:

- **Time = propagation delay**: Time is not an intrinsic parameter of change, but an emergent effect of information transfer across space.
- **Zero-distance limit**: As $\Phi \to 0, \mathbb{D} \to 0$, no propagation is required, hence $\tau \to 0$, corresponding to a timeless overlapping state.
- **Scale invariance (origin of light speed)**: Within any local region of fixed Φ, the ratio between $\mathcal{C}(\Phi)$ and $\mathbb{D}$ remains constant, producing the observed invariance of the time–space conversion rate (effective speed of light).
- **Cross-layer effects**: When observations involve regions with different Φ, variations in $\mathcal{C}(\Phi)$ lead to rescaling of time flow and distance perception, corresponding to relativistic phenomena.

Structural Embedding Example

To demonstrate the **embedability** of the SIC framework, this section provides a minimal example.
This does not aim to replace or optimize existing theories, but rather to illustrate how established physical descriptions can be reformulated within the SIE structure.
In the regime of macroscopically stable expansion ($\Phi \approx 1$), the spatial expansion rate approaches a constant. Under this condition, SIE-1 and SIE-2 can be approximated as fixed background parameters. The system's dynamics can then be described by local structural variations, without explicit dependence on the global expansion parameter.
Within this limit, conventional descriptions of motion based on distance ($\mathbb{D}$) and time (τ) emerge as effective approximations of the SIE framework. In other words, the equations of motion in classical mechanics can be interpreted as local dynamical behavior under a fixed-Φ condition.
This result implies that existing physical theories can be understood as **effective projections** of SIC under specific conditions, rather than independent foundational structures.

Failure Criterion

The validity of SIC is grounded in its **structural consistency** and **embedability**. To ensure falsifiability, the framework explicitly defines its failure conditions as follows:
SIC is considered invalid if there exists a physical description that simultaneously satisfies:

- It corresponds to stable and reproducible experimental observations
- It forms an internally self-consistent theoretical structure with predictive power
- It cannot, under any variable redefinition or limiting condition, be reformulated into a form consistent with the SIE equations

In the third condition, "embedding" is defined as the existence of a mapping of variables and parameter transformations such that the core relations of the theory can be expressed as a **special case, limit, or approximation** of the SIE framework.
This criterion ensures that SIC is not a closed system, but an open structure subject to challenge and potential falsification by external theories.

> In SIC, "division by zero" is not treated as a mathematical operation, but is reinterpreted as a **boundary condition of geometric states**.
> As the system approaches the limit $\Phi \to 0$, all conventional definitions break down, and physical quantities degenerate into an indeterminate overlapping state.
> This state corresponds to maximal information potential and an unexpanded geometric configuration.

I. REDEFINITION OF VELOCITY: INFORMATION PROPAGATION DELAY

> **Within the SIC framework, velocity is not an independent physical quantity of motion, but rather the geometric ratio of delay associated with information propagation in an already expanded spatial structure.**

In conventional physics, velocity is defined as the rate of displacement with respect to a fixed background of space and time. This assumption leads to increasingly complex formulations such as spacetime curvature. SIC provides a deeper reinterpretation at the structural level:

Propagation Delay Interpretation

What is commonly described as "motion" is not the change of position over time, but the process by which **information (Ξ)** is transmitted and reconstructed across interlayers defined by different spatial expansion rates (Φ).
This propagation requires traversing geometrically induced separation distances ($\mathbb{D}$), which arise from information decoupling, and incurs a corresponding delay cost (τ).Therefore, velocity is not a fundamental quantity, but rather the **observational projection** of the ratio between distance ($\mathbb{D}$) and delay (τ) within a macroscopic interlayer.

Geometric Origin of the Invariance of Light Speed

Light (electromagnetic waves), as the most fundamental mode of information propagation within an interlayer, is not defined by its numerical speed, but by the fact that it maintains a **constant conversion ratio between space and time**.Within any locally stable interlayer (fixed Φ), the delay cost (τ) required for information to traverse a distance ($\mathbb{D}$) maintains a stable proportionality to that distance. This gives rise to the observed propagation limit c.
Thus, the invariance of the speed of light does not arise from any intrinsic absoluteness of light itself, but from the fact that all observational processes are inherently constrained by the same local space–time conversion ratio.In other words, both the observer and the measurement standard are simultaneously governed by this ratio.

Origin of Nonlocality in Microscopic / High-Energy Regimes

In microscopic interlayers or extreme high-energy regimes ($\Phi \to 0$), space and information remain in a highly overlapping state, and spatial separation has not fully emerged ($\mathbb{D} \to 0$).Under these conditions, information propagation does not require traversal across geometric distance, and the associated delay cost (τ) approaches zero.
The observed phenomena of instantaneous correlation or long-range synchronization do not imply superluminal transmission, but instead reflect the absence of spatial separation at the underlying geometric level.

Therefore, quantum entanglement and wave-like distributions do not violate the speed-of-light constraint. Rather, they represent synchronized manifestations of information within a low-expansion regime.Causality, in this context, is not propagated along a temporal sequence, but is determined by the **global consistency of the geometric configuration**.

II. DISCRETE SETTLEMENT OPERATOR: GEOMETRIC TRANSITIONS REPLACING CONTINUOUS EVOLUTION

Within the SIC framework, the universe does not evolve through continuous "flow", but rather through a sequence of **discrete settlement events** that update system states.

When $\Phi \to 0$ or when local geometric conditions undergo significant variation, the conventional differential description $\frac{d}{d\tau}$ loses its underlying physical meaning.

> **SIC postulates that physical state changes do not occur through smooth temporal evolution, but through instantaneous settlement events triggered when information propagation and geometric configurations satisfy specific conditions. Time is merely the manifestation of delay between such settlement events.**

1. Interlayer Settlement Operator

$$\mathbf{S}[\Xi, \Phi] \cdot \Psi_n = \Psi_{n+1} \tag{6}$$

$$\Psi_{n+1} = \mathrm{Ker}(\mathbf{S}) \oplus \Delta\Xi_{\text{quantized}} \tag{7}$$

Interpretation:

- **Operator S:**
  The settlement operator does not track temporal evolution. Instead, it continuously evaluates whether local geometric conditions—defined by information density (Ξ), expansion rate (Φ), and structural constraints—satisfy the criteria for settlement.

- **Discrete Transition:**
  The transition from state n to $n+1$ does not occur along a continuous trajectory. Rather, it is triggered instantaneously once settlement conditions are fulfilled.
  This process is governed not by time evolution, but by the **validity of geometric and informational configurations**.

- **Relation to Time:**
  Between two settlement events, the system does not "evolve" in the conventional sense. Instead, information has not yet completed propagation across interlayer distance ($\mathbb{D}$) nor paid the required delay cost (τ).
  Therefore, time represents the **waiting cost between settlements**, rather than the driving mechanism of state change.

2. Falsifiability and Empirical Predictions of Discrete Settlement

If the underlying operation of the universe is governed by discrete settlement, then at sufficiently high observational resolution, one should detect a **breakdown of continuous evolution**.

[Test of Discreteness]

If, in attosecond-scale (or higher-resolution) experiments, physical processes—such as electron ionization or energy level transitions—are observed to exhibit fully continuous and arbitrarily divisible trajectories, with no indication of discrete transitions or settlement delays, then the discrete settlement mechanism proposed by SIC would be falsified.

III. GEOMETRIC CAUSALITY MATRIX: THE LOGIC OF DISCRETE SETTLEMENT

Building upon the previously defined **Discrete Settlement Operator**, SIC further formalizes physical phenomena as conditions of **geometric admissibility**.

Within this framework, causality is no longer interpreted as a chain of events along a temporal axis, but as a criterion of whether local interlayer configurations satisfy the conditions required for settlement.

Geometric admissibility precedes physical realization; time is merely the delay between settlement events.

This implies that the perceived ordering of events does not originate from time itself, but emerges from the sequencing of multiple discrete settlement points under information propagation delay (τ).

To quantify this structure, SIC defines the core settlement matrix $\mathbb{M}_{\text{settle}}$:

$$\mathbb{M}_{\text{settle}} = \begin{bmatrix} \Phi & \Xi \\ \nabla\Phi & \vec{\omega} \end{bmatrix} \quad , \quad \det(\mathbb{M}_{\text{settle}}) = \Phi \cdot \vec{\omega} - \Xi \cdot \nabla\Phi \geq \text{Threshold} \tag{8}$$

Matrix Variable Interpretation:

- **Φ (Expansion Term):**
  Represents the geometric openness of space, determining whether information can separate and form observable structures.
- **Ξ (Information Term):**
  The density of information carried within the interlayer, serving as the globally conserved core quantity.
- **$\nabla\Phi$ (Gradient Term):**
  The geometric tension arising from non-uniform spatial expansion, corresponding to gravitational and curvature effects.
- **$\vec{\omega}$ (Rotational Component):**
  The topological dynamical structure formed to maintain local stability and conservation of information (e.g., spin, magnetic field).

1. Settlement Trigger Mechanism

Within any local interlayer, the four variables form a **closed, coupled system**. The system continuously evaluates geometric admissibility, yet this process **does not correspond to the passage of time**:

- **Static Undetermined State:**
  When
  $$\det(\mathbb{M}_{\text{settle}}) < \text{Threshold} \tag{9}$$
  the region remains in a *non-settleable state*. Although local variations in information may exist, the conditions required for cross-distance transmission ($\mathbb{D}$) and delayed manifestation (τ) are not satisfied. As a result, no observable event is formed.
- **Dynamic Settlement Point:**
  When information accumulation ($\Xi \uparrow$), spatial compression ($\nabla\Phi \uparrow$), or rotational compensation (adjustments in $\vec{\omega}$) drives the determinant beyond the critical threshold, the system undergoes a **forced discrete settlement**.

  At this point, information completes its interlayer transfer and pays the required delay cost (τ), thereby manifesting as an observable event in the macroscopic layer.

2. Emergence of Temporal Ordering

Since each settlement requires a finite information transmission delay, the "flow of time" perceived by an observer is not fundamentally continuous. Instead, it emerges as an **ordering of discrete settlement events**, structured by their associated delay costs (τ).

Time is not the cause of events, but the manifestation of delay structures between events.

In macroscopically stable regions ($\Phi \approx 1$), where delay costs are approximately linear and uniform, discrete settlements statistically approximate a continuous temporal flow.

This gives rise to the conventional treatment of time as a continuous parameter in classical physics.

3. Physical Consequences of Geometric Causality

By reformulating causality as a **matrix-based settlement condition**, SIC provides a unified interpretation of several key physical phenomena:

- **Instantaneity of Quantum Transitions:**
  Electrons do not traverse continuous trajectories between states.
  Instead, transitions occur only when the matrix condition is satisfied, resulting in an immediate settlement.
  The intermediate path does not exist geometrically, and thus requires no time.

- **Coupling of Gravity and Fields:**
  When $\nabla\Phi$ increases (spatial compression), the system must redistribute $\vec{\omega}$ or Ξ to maintain determinant balance.
  This corresponds to the emergence of strong magnetic fields and extreme energy densities in astrophysical systems.

- **Origin of Observational Limits:**
  All observations rely on information transmission across $\mathbb{D}$ and the associated delay cost τ.
  Therefore, observers can never directly access the **pre-settlement state**, establishing a fundamental limitation on physical measurement.

IV. DYNAMIC ALLOCATION OF GEOMETRIC ADDRESSING AND TEMPORAL SETTLEMENT COST

From the Operator $\hat{O}$ to the Quantum Rollback Mechanism

1. How Observation Forces $\Phi \to 1$ (Projection Locking)

In SIC, unobserved information exists in an **overlapped/interlayer state**, where $\Phi < 1$.
This indicates that geometric resolution is not yet saturated, and the system remains in an allocatable configuration.

> **The observation operator $\hat{O}$ is not a passive measurement, but an "Information Synchronization Request," forcing the system to complete geometric mapping and settlement of information that is still in delayed transmission.**

$$\hat{O}\,|\Psi\rangle_{\Phi<1} \xrightarrow{\text{Register Access}} \mathbb{M}_{\text{settle}} \;\Rightarrow\; \Phi \to 1 \tag{10}$$

Mechanism:

- **Geometric Mapping Request:**
  Observation demands that a segment of information Ξ, originally in a delayed transmission state, be mapped onto a concrete geometric address.

- **Address Allocation:**
  To fulfill this mapping, the system terminates the ongoing delay chain and compresses the information into a single geometric location, forcing $\Phi \to 1$ and completing manifestation.

- **Nature of Settlement:**
  This process is fundamentally a **forced settlement of accumulated delay**.
  Distributed propagation is interrupted, and its geometric cost (τ) is paid instantaneously.

- **Exclusivity Constraint:**
  Once an address is allocated, it cannot host multiple incompatible information states.
  The system must perform a **selection-based settlement**, corresponding to quantum collapse.

> **Observation is not "seeing a result," but a forced settlement and geometric write operation applied to information still in delayed transmission.**

2. Influence of Observation on Time Generation (SIE-3)

In SIC, time (τ) is defined as the geometric delay cost incurred during information transmission across expanded space. Therefore, observation does not merely read information—it **modifies the delay structure itself**.

$$\Delta\tau_{\hat{O}} = \int \frac{\partial \Phi}{\partial \Xi} \cdot \hat{O}(\Delta\Xi)\, d\mathbb{D} \tag{11}$$

Interpretation:

- **Interruption of Delay:**
  Normally, information propagates across $\mathbb{D}$ while accumulating delay (τ).
  Observation interrupts this process, concentrating the delay into a single settlement.
- **Acceleration of Settlement:**
  Propagation shifts from gradual delay accumulation to instantaneous settlement, effectively compressing the local time-generation process.
- **Redistribution of Time:**
  To preserve global consistency, the system redistributes delay cost across other regions, maintaining overall geometric conservation.

  Observation does not create time—it redistributes the structure of delay.

3. Erasure Operator $\hat{E}$: Geometric De-allocation and Rollback Mechanism

Corresponding to the observation operator $\hat{O}$, SIC defines a reverse operation: the erasure operator $\hat{E}$.
Its function is not to eliminate information, but to **remove geometric address allocation**.

$$\hat{E}\left(\hat{O}|\Psi\rangle_{\Phi=1}\right) \xrightarrow{\text{De-allocation}} |\Psi\rangle_{\Phi<1} \tag{12}$$

Mechanism:

- **Address Release:**
  The erasure operation releases previously assigned geometric addresses, detaching information from fixed spatial positions.
- **Return to Delay State:**
  Once the address is removed, information re-enters the interlayer transmission structure, transitioning from manifestation ($\Phi = 1$) back to delayed propagation ($\Phi < 1$).
- **Non-destructive Nature:**
  Information is not destroyed; it simply loses the conditions required for manifestation and becomes unobservable.

  Erasure does not eliminate information—it returns it to a pre-settlement, delayed transmission state.

V. SIE REFORMULATION OF MACROSCOPIC PHYSICAL EQUATIONS

By introducing the spatial expansion rate Φ and the delay structure of information propagation, existing macroscopic physical equations can be reformulated at a deeper level.
SIC does not reject established equations, but rather identifies them as **approximate results valid under specific expansion conditions and stable delay structures**.

1. Temporal Genesis Equation

$$\tau = \int \Phi \cdot d\mathbb{D} \tag{13}$$

Interpretation:

- **Delay as Time:**
  Time (τ) is not a background dimension, but the accumulated propagation delay incurred as information traverses geometric distance ($\mathbb{D}$) within expanded space.
- **Non-generative Nature:**
  Time is not "produced" by space, but emerges as an unavoidable consequence of information transmission within already expanded geometry.
- **Limiting Case:**
  When $\Phi \to 0$ or $\mathbb{D} \to 0$, no geometric path is required for transmission, and delay vanishes ($\tau \to 0$), corresponding to a nonlocal synchronous state.

  Time is not a driving parameter of the system, but the cost incurred when information cannot be transmitted instantaneously.

2. Mass–Energy Symmetry and Information Density Correction

$$\Xi = \frac{E}{\Phi} = m \cdot c^2 \tag{14}$$

Interpretation:

- **Primacy of Information:**
  Energy and mass are not fundamental quantities, but manifestations of information density (Ξ) under different expansion conditions.
- **Expansion Modulation:**
  In low-expansion regions ($\Phi < 1$), the same information corresponds to higher energy density;
  in high-expansion regions ($\Phi \to 1$), information stabilizes into mass-like structures.
- **Consistency with Standard Physics:**
  When $\Phi = 1$, the relation reduces to the standard expression $E = mc^2$, indicating that conventional physics represents a limiting case of full expansion.

3. Interlayer Propagation and Cross-Domain Observation

$$\nabla^2 \mathbf{A} - \frac{1}{c^2\Phi^2}\frac{\partial^2 \mathbf{A}}{\partial \tau^2} = 0 \tag{15}$$

Interpretation:

- **Local Invariance:**
  Within a single interlayer (fixed Φ), the observed speed of light c remains constant, reflecting the intrinsic proportionality of the local delay structure.

- **Ratio-Based Nature:**
  The invariance of light speed does not arise from an absolute velocity, but from a fixed ratio between temporal delay and spatial scale.

- **Cross-domain Effects:**
  When observation spans regions with varying expansion rates ($\nabla\Phi \neq 0$), information propagation encounters different delay structures, resulting in frequency shifts and energy redistribution (e.g., redshift).

  The constancy of light speed arises from the locking of the time–space ratio within a local observational frame, not from an absolute propagation speed.

4. Geometric Tension Equation

$$G_{\mu\nu} + \Lambda g_{\mu\nu} \Rightarrow \kappa \cdot \nabla\Phi \tag{16}$$

Interpretation:

- **Redefinition of Gravity:**
  Gravity is not the curvature of spacetime caused by mass, but the distribution of geometric tension arising from spatial expansion gradients ($\nabla\Phi$).

- **Distortion of Delay Structure:**
  In regions of high $\nabla\Phi$, information propagation paths are reconfigured, increasing delay cost and manifesting as time dilation and trajectory bending.

- **Reinterpretation of Dark Components:**
  Dark matter and dark energy may be understood as effective phenomena arising from unmodeled Φ gradient distributions, rather than independent physical entities.

  Gravity is not a force, but the inhomogeneous distribution of information propagation delay within space.

VI. SIE REFORMULATION OF MICROSCOPIC QUANTUM EQUATIONS

Within the SIC framework, non-intuitive quantum phenomena—such as wave–particle duality, superposition, and entanglement—are not fundamental uncertainties.
Instead, they arise as **projections of information propagation under different spatial expansion conditions and delay structures**.

By incorporating the spatial expansion rate (Φ) and the delay–settlement mechanism into quantum operators, SIC unifies "wave," "particle," and "measurement" within a single geometric framework, without requiring an additional collapse postulate.

1. Geometric Reformulation of the Schrödinger Equation

$$i\hbar \frac{\partial}{\partial \tau}\Psi = \left[-\frac{\hbar^2}{2m\Phi}\nabla^2 + V(\Phi)\right]\Psi \tag{17}$$

Interpretation:

- **Nature of Evolution:**
  Wavefunction evolution does not occur "in time," but reflects the accumulation of propagation delay as information traverses the interlayer structure.

- **Expansion Modulation:**
  Φ determines spatial resolvability.
  When $\Phi < 1$, information remains in an overlapped propagation state, giving rise to wave-like behavior.

- **Limiting Behavior:**
  As $\Phi \to 1$, the delay structure stabilizes, and the equation reduces to classical particle dynamics.
  As $\Phi \to 0$, spatial separation vanishes, and information becomes globally overlapped.

  The wavefunction is not a description of a particle, but the propagation structure of information prior to delay settlement.

2. Geometric Origin of Wave–Particle Duality

In SIC, wave-like and particle-like behaviors are not complementary properties, but correspond to whether **delay settlement has occurred**:

- **Wave-like (Pre-settlement State):**
  When $\Phi < 1$, information is still propagating within the interlayer, and delay has not yet been concentrated.
  This manifests as a spatially distributed probability structure.

- **Particle-like (Post-settlement State):**
  When observation occurs ($\hat{O}$ acts), delay is forcibly settled, and information is locked into a single geometric address, appearing as a particle.

  Wave and particle are not distinct entities, but different manifestations of the same information under propagation and settlement.

3. Reconstruction of Quantum Entanglement

In conventional interpretations, entanglement appears to involve superluminal influence.
In SIC, this phenomenon arises because the information has not yet formed geometric separation ($\mathbb{D} \to 0$).

- **Shared Delay Structure:**
  Entangled particles do not influence each other through space, but share the same un-settled information structure.

- **Synchronous Settlement:**
  When one side is observed, the entire shared structure undergoes simultaneous forced settlement, producing instant correlation.

- **No Superluminal Transfer:**
  No information propagates across space; therefore, no violation of light-speed constraints occurs.

  Entanglement is not "action at a distance," but the simultaneous settlement of a single, non-separated structure.

4. Uncertainty Principle as Delay Structure

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2\Phi} \tag{18}$$

Interpretation:

- **Delay-Induced Indeterminacy:**
  When $\Phi < 1$, information has not completed propagation settlement, making position and momentum simultaneously undefined.

- **Observation Constraint:**
  When observation forces $\Phi \to 1$, information becomes localized, reducing uncertainty but disrupting the original propagation structure.

- **Consistency Limit:**
  In the macroscopic limit ($\Phi \to 1$), this relation reduces to the standard uncertainty principle.

  Uncertainty is not a limitation of measurement, but a geometric consequence of incomplete delay settlement.

5. Quantum Measurement and Rollback Mechanism ($\hat{O}$ and $\hat{E}$)

In SIC, the quantum measurement problem can be fully described by the mechanisms of **delay settlement** and **geometric de-allocation**, without requiring additional interpretations.

- **Observation Operator $\hat{O}$:**
  Forces distributed information within the delay structure to settle and map onto a geometric address ($\Phi \to 1$).

- **Erasure Operator $\hat{E}$:**
  Removes the assigned geometric address, returning information to a pre-settlement propagation state ($\Phi < 1$).

- **Rollback:**
  Once the address is released, information re-enters the delay propagation structure and resumes wave-like behavior.

  The "quantum measurement problem" is fundamentally the transition between delayed propagation and forced settlement of information.

VII. SPATIAL EXPANSION AND HEAT DISSIPATION UNDER INFORMATION CONSERVATION

(SIC Framework Interpretation)
Core Principle:
In conventional physics, energy conservation is regarded as a fundamental law of the universe.
However, within the SIC framework, a crucial distinction must be made between two structural layers:

- **Background Interlayer (Unmanifested):** The only strictly conserved quantity is **information density** (Ξ)

- **Manifested Reality (Observable Layer):** Energy is a **geometric projection** arising from the action of the projection operator $\mathcal{P}_\beta$

Therefore, energy is not a globally fundamental conserved quantity, but rather a **locally conserved manifestation** of information under specific geometric conditions.
Within this framework, the release, transformation, and redistribution of energy must be realized through a consistent geometric carrying mechanism.
Spatial expansion is precisely the geometric response to such information redistribution.

> **What is traditionally referred to as "waste heat" and entropy increase is not the dissipation of energy, but the reallocation of information that can no longer sustain high-density manifested structures, thereby driving geometric expansion of space.**

1. Dimensional Reduction of Conservation and Geometric Reformulation of the Second Law

$$\oint d\Xi = 0 \quad \text{(Global Information Conservation in Interlayer)} \tag{19}$$

$$\sum E_{\text{obs}} = f\big(\mathcal{P}_\beta(\Xi)\big) \quad \text{(Local Energy Conservation in Manifested Reality)} \tag{20}$$

Interpretation:

- **Unmanifested State (Background Layer):**
  The total information Ξ is strictly conserved and does not take the form of energy.

- **Manifested State (Observable Layer):**
  Energy conservation emerges as a **boundary condition**, resulting from the projection of information via $\mathcal{P}_\beta$ onto finite geometric structures (Σ, L), rather than a fundamental law.

- **Entropy Increase ($\Delta S \geq 0$):**
  When local information density exceeds the carrying capacity of the geometric structure, the system must reduce the density to maintain consistency.
  This process manifests as:
  - Information that cannot be organized into structured states $\rightarrow$ released as "waste heat"
  - Increased geometric demand $\rightarrow$ triggers expansion of spatial volume

  Thus, entropy increase is not a growth of disorder, but a geometric redistribution of information required to maintain manifestability.

VIII. DYNAMIC COUPLING BETWEEN MICRO AND MACRO SCALES: THE INFORMATION–SCALE PROPORTIONALITY HYPOTHESIS

This section establishes that the distinction between microscopic and macroscopic regimes is not a fundamental physical separation, but rather a dynamic geometric phase transition.
Within the SIC framework, the "microscopic" domain is not defined by a fixed scale, but emerges as a boundary determined jointly by information density and spatial expansion conditions.

1. Information–Scale Proportionality Equation

$$R_{\text{boundary}} = \eta \cdot \Xi_{\text{overlap}} \tag{21}$$

Core Hypothesis:
The effective spatial influence range of a microscopic (overlap-state) configuration is proportional to its internal **information/energy overlap** Ξ.
This proportionality defines the boundary scale at which microscopic properties become manifest at macroscopic levels.

- **Geometric Expansion Interpretation:**
  In conventional physics, high energy is typically associated with small spatial scales.
  In the SIC framework, this is reinterpreted as high information density extending the influence range of low-Φ structures outward, rather than simply "compressing space."
- **Black Hole Interpretation:**
  When information accumulation reaches extreme levels, structural features originally confined to microscopic regimes can extend their influence to macroscopic scales.
  The event horizon may thus be understood as a stable boundary of such high-information-density regions within the macroscopic interlayer, rather than a strictly physical surface.

2. Interlayer Structural Strength and Phase Transition Threshold

$$\nabla^2 \Phi = \kappa (T_{\mu\nu} - S_{\text{interaction}}) \tag{22}$$

Mathematical Interpretation:
$S_{\text{interaction}}$ represents the structural stabilization term arising from fundamental interactions, characterizing the ability of space to maintain geometric separation across different scales.

- **Stability Constraint:**
  There exists a set of stable minimal structural scales in the universe, which can be interpreted as lower-bound constraints imposed by interactions on spatial expansion, preventing arbitrary collapse of Φ.
- **Expansion Phase Transition:**
  When the local energy density $T_{\mu\nu}$ exceeds the structural support capacity, Φ may transition into a new geometric regime.
  As a result, structural characteristics that were previously confined to local (microscopic) domains can emerge at larger scales, producing a phenomenon akin to "microscopic leakage" into the macroscopic regime.

IX. EMERGENCE OF MATTER AND THE MECHANISM OF ROTATIONAL LOCKING

Within the SIC framework, matter and energy are not treated as primitive entities, but as geometric outcomes of information undergoing spatial expansion.
In the extreme microscopic limit ($\Phi \to 0$), information (Ξ) lacks separable structure and temporal ordering, forming a highly overlapped global state.
As spatial expansion proceeds ($\Phi \uparrow$), information must adopt specific topological configurations to maintain stability, which can be categorized into open and closed structures.

1. The Nature of Energy: Open Spiral Projection

$$E_{\text{open}} = \int \Xi \left(\vec{\omega}_{\text{open}} \cdot d\vec{\mathbb{D}} \right) \tag{23}$$

Interpretation:

- **Propagation as Openness:**
  When information does not form a closed structure, it manifests as open rotational or wave-like modes in space.
  This behavior corresponds to what is observed as "energy propagation."

- **Massless Character:**
  Due to the absence of closure and settlement conditions, such information flows do not form stable localized structures, and therefore do not correspond to rest mass (e.g., photons).

2. Rotational Locking and Mass Emergence

$$m_{\text{locked}} = \frac{1}{\Phi c^2} \oint_C \Xi \left(\vec{\omega}_{\text{closed}} \times \vec{\mathbb{D}} \right) \cdot d\vec{s} \tag{24}$$

Interpretation:

- **Rotation as Localization:**
  When initially open information flow is constrained into a closed loop by geometric conditions, its dynamics become confined within a localized region.

- **Emergence of Mass:**
  Such closed configurations manifest in the macroscopic interlayer as stable entities.
  Mass is therefore not a fundamental property, but a form of **stable topological confinement of information**.

3. The Nature of Gravity: Compensatory Vortices from Information Conservation

$$\nabla \Phi_{\text{gravity}} = -\kappa \left(\frac{\partial \vec{\omega}_{\text{closed}}}{\partial \tau} \times \Xi_{\text{locked}} \right) \tag{25}$$

Interpretation:

- **Geometric Tension:**
  When information becomes locally locked (i.e., forms mass), it disrupts the uniformity of spatial expansion.

- **Spatial Response:**
  Surrounding space generates compensatory gradients to restore geometric consistency, which are macroscopically observed as gravity.

4. Topological Constraints and Mass Singularities: A Geometric Necessity

> **Mass may be understood as the stable manifestation of topological singularities arising when rotational structures cannot be smoothly extended within the expanded interlayer.**

When information attempts to form closed rotational structures in three-dimensional space, it is constrained by fundamental geometric theorems (e.g., the hairy ball theorem):

- **Emergence of Singularities:**
  On closed topologies (such as S^2), any continuous vector field must contain zeros.
  These points of discontinuity correspond to geometric singularities and act as anchors for stable structures.

- **Magnetic Constraints:**
  To maintain closed circulation, the rotational structure must satisfy a source-free condition
  ($\oint \vec{B} \cdot d\vec{A} = 0$), consistent with the fundamental properties of magnetic fields.

$$\sum_i \mathrm{ind}_{x_i}(V) = \chi(S^2) = 2 \tag{26}$$

Conclusion:
There exists an intrinsic structural coupling between information distribution and spatial topology.
Matter is not arbitrarily formed, but emerges as a necessary projection of geometry in order to complete **closed settlement**.

X. LOGICAL PHASE TRANSITION: BEYOND THE ILLUSION OF CONTINUITY AND THE SECOND MATHEMATICAL CRISIS

This section identifies a long-standing tension in modern physics between mathematical formalism and physical reality.
On one hand, the existence of minimal resolvable scales (e.g., the Planck scale) is widely acknowledged.
On the other hand, continuous limits remain the dominant foundation for theoretical description.
While this coexistence is effective in many regimes, it reveals structural limitations under extreme conditions.

$$\text{Paradox: } \begin{cases} \text{Observation: } \Delta x \geq l_P,\ \Delta t \geq t_P \\ \text{Calculus: } \lim_{\Delta x \to 0} f(x) \end{cases} \Rightarrow \text{Domain Mismatch} \tag{27}$$

Core Tension

When a physical system possesses a minimal resolution scale, the operation of taking limits "toward zero" no longer corresponds to an actual physical process, but instead becomes an effective approximation tool.
When such approximations are extended beyond their domain of validity, they give rise to divergences and singularities.

1. Physical Manifestation of the Second Mathematical Crisis

Modern mathematics established the rigor of continuous systems through formal limit definitions.
However, in physical applications where the underlying structure exhibits discreteness or threshold behavior, the domain of applicability of these tools must be re-evaluated.
Certain renormalization procedures may be interpreted as compensatory mechanisms addressing this mismatch.

2. Geometric Exclusivity and Overlap Conditions

In a purely continuous and infinitely divisible space, distinct states are strictly separable.
However, within the SIC framework, when geometric resolution falls below a certain threshold, this separability breaks down, and information enters an overlapping state ($\Phi \to 0$).
This does not violate continuity, but rather indicates that continuity itself has a bounded domain of applicability.

3. Settlement Logic and Continuous Approximation

At the fundamental level, physical processes may be represented as discrete state transitions, while continuous descriptions emerge as effective limits under high-frequency settlement.

Within SIC, certain "singularities" are reinterpreted not as failures of theory, but as boundaries of geometric or logical phase transitions.

XI. GEOMETRIC REDEFINITION OF FERMIONS AND BOSONS: MANIFEST ENTITIES AND INTERLAYER MEDIATORS

Structural Perspective:
The Standard Model classifies particles based on spin.
SIC provides a complementary geometric interpretation:
fermions are **stable projections of information within the macroscopic interlayer**, whereas bosons are **dynamic transmission modes of information across different expansion layers**.

1. Fermions: Macroscopic Manifestation of Microscopic Information and the Nature of Energy Levels

$$\Psi_{\text{Fermion}} = \mathbf{P}_{\Phi\to 1}\left[\oint \Xi(\vec{\omega})d\tau\right] \tag{28}$$

$$[\Delta]_{\Xi} \Rightarrow \text{Non-manifested State} \Rightarrow \text{New Projection} \tag{29}$$

Interpretation:

- **Manifested Structure:**
  Fermions arise when information satisfies geometric stability conditions and becomes localized within the $\Phi = 1$ layer.
  This form of **geometric occupancy** gives rise to physical exclusivity (e.g., Pauli exclusion).
- **Energy Level Interpretation:**
  Energy levels are not orbital assumptions, but discrete solutions of geometric stability conditions.
- **Transition Mechanism:**
  During transitions, the system temporarily exits the manifested condition ($\Phi < 1$).
  This explains the absence of intermediate trajectories in microscopic transitions:
  the system is effectively "withdrawn" from three-dimensional geometry during the process.

2. Bosons: Interlayer Information Transmission Modes

$$\mathbf{B}_{\text{Boson}} = \frac{\partial \Xi}{\partial \Phi} \cdot e^{i(\vec{k}\cdot\vec{\mathbb{D}} - \omega\tau)} \tag{30}$$

$$\lim_{\Phi\to 0} \mathbf{B}_{\text{Boson}} \equiv \Xi_{\text{pure}} \tag{31}$$

Interpretation:

- **Dynamic Information States:**
  Bosons represent propagating disturbances of information across expansion gradients (Φ), rather than entities with fixed geometric boundaries.
- **Interlayer Mediation:**
  They serve as the transmission mechanism through which the system performs state reconfiguration and settles associated energy costs.
- **Interpretation of Light Speed:**
  The speed of light c corresponds to the geometric limit of **settled information transmission** within the expansion structure, rather than a purely dynamical constant.

  Conclusion:
  Fermions correspond to **outcomes (stable manifestations)**,
  whereas bosons correspond to **processes (transmission mechanisms)**.
  At the geometric level, they do not belong to the same ontological category.

XII. GEOMETRIC INTERPRETATION OF GRAVITATIONAL WAVES: FROM FIELD PERTURBATIONS TO STRUCTURAL UPDATES

Core Perspective:
In standard theory, gravitational waves are described as propagating perturbations of spacetime curvature.
Within the SIC framework, they are reinterpreted as a **dynamic refresh process of spatial geometry under information reconfiguration**.
The observed effect is not a force transmission, but a temporary reorganization of local geometric scales.

1. Geometric Dynamics Correspondence

Stable structures are governed by gradients of Φ, while dynamical disturbances follow a geometric wave relation:

$$\vec{g} = \nabla\Phi_{\text{effective}} \tag{32}$$

$$\Box\Phi = \zeta_{\Xi} \cdot \frac{\partial^2 \Xi}{\partial \tau^2} \tag{33}$$

Interpretation:

- **$\vec{g}$:**
  Represents stable geometric tension (gravity).
- **$\Box\Phi$:**
  Describes fluctuations in spatial expansion rate induced by rapid variations in information density Ξ.
  This corresponds to a large-scale **geometric reconfiguration process**, where information flow is redistributed rather than "emitted" or "lost."

2. Observational Interpretation: Temporary Reconfiguration of Measurement Scales

When a geometric disturbance (gravitational wave) passes through an observation region, it alters local metric relations, leading to measurable deviations:

- **Scale variation:**
  Appears as transient changes in interferometer arm lengths, reflecting local fluctuations in Φ.
- **Propagation limit:**
  The spread of this structural update is constrained by the geometric limit of information settlement (identified as c).

$$\Delta L = L \cdot \int_{\text{pulse}} \delta\Phi(t)\, dt \tag{34}$$

Clarification:
The quantity ΔL does **not** represent mechanical deformation caused by external forces.
Instead, it reflects a **redefinition of measurement scale** due to local variations in geometric structure (Φ).

Conclusion:
Gravitational waves are not transmissions of force, but **dynamic updates of geometric nodes**.
What LIGO detects is not spacetime "vibration," but a ripple of **geometric redistribution driven by information reorganization**.

3. Information Saturation Limit and Discrete Spatial Scale

Based on the principle of entropy saturation at the black hole horizon (normalized to unity), and under a non-zero displacement loss β, the total information capacity of a single discrete spatial unit must satisfy a convergence condition:

Information Saturation Limit Σ

$$\Sigma = \frac{1}{1-\beta} \times \mathcal{C}_{\text{geom}} \approx 11.56 \tag{35}$$

Corresponding fundamental spatial resolution:

$$L = \sqrt{\Sigma} = 3.4 \tag{36}$$

Note:
This result is **not an empirical fit**, but a consequence enforced by geometric convergence conditions.
Implication:
The value 3.4 is not accidental.
It reflects an intrinsic loss factor in geometric mapping, indicating that spacetime is not infinitely divisible, but instead discretized into fundamental units constrained by Σ.

CORE DERIVATION: FROM BOUNDARY GEOMETRY TO SYSTEMIC DISPLACEMENT

The original motivation of this work was to resolve the rigidity problem of information propagation at the **Cosmic Information Boundary**, particularly in the presence of early-formed galaxies.

Conventional cosmology assumes symmetric expansion. However, when evaluating boundary information loading rates, divergence terms emerge.
To preserve global continuity of information flow, it becomes necessary to introduce an **interlayer displacement compensation mechanism**.

1. Boundary Flux Equilibrium and the Divergence Problem

In the initial formulation, the universe is treated as a finite manifested system subject to boundary constraints on information flux:

$$\oint_{\partial\Sigma} \left(\nabla\Phi \cdot \vec{n} + \vec{\beta}_{\text{sys}} \right) dA = 0 \tag{37}$$

where:

- $\nabla\Phi \cdot \vec{n}$: information flux induced by spatial expansion
- $\vec{\beta}_{\text{sys}}$: **systemic displacement compensation term** required to maintain flux balance

Key Result:
If $\vec{\beta}_{\text{sys}} = 0$, divergence inevitably emerges in regions of high information density, as the flux term cannot be globally neutralized.

> **Therefore, to preserve global continuity of information flow, the system must spontaneously generate a non-zero displacement compensation.**

2. From Boundary Compensation to a Systemic Axis

When the compensation term is propagated back from boundary conditions into the global geometric structure, the assumption of a perfectly symmetric (isotropic) universe becomes inconsistent.

Mathematically, the system is forced to select a stable displacement direction that minimizes overall geometric tension.

Geometric Constraint and Structural Assumption

To further resolve the numerical origin of this configuration, we introduce a structural assumption:

> The spatial interlayer, in the microscopic limit, tends toward configurations that maximize local packing density.

In three-dimensional space, this corresponds to **close-packing geometry**, where the topology is constrained by the **kissing number**, limiting each node to at most 12 stable contacts.

Minimal Standing Wave Constraint

Within this 12-node interference network, any global compensation displacement $\vec{\beta}_{\text{sys}}$ must embed into the topology in a manner that minimizes geometric tension.

This condition is equivalent to:

> Identifying the shortest closed standing-wave path that preserves phase coherence across a finite node network.

Under this constraint, the dimensionless minimal path length (i.e., displacement ratio) is geometrically bounded within:

$$|\vec{\beta}| \sim 0.08 \, - \, 0.09 \tag{38}$$

Numerical Convergence and Parameter Identification

To further constrain this value, we apply a multi-source alignment approach, incorporating:

- dimensionless structures among known physical constants
- cosmological observations (e.g., Hubble parameter discrepancies)
- numerical simulations under geometric constraints

High-dimensional feature matching and parameter space exploration are then performed to identify a stable solution, yielding:

$$|\vec{\beta}| \approx 0.0865 \tag{39}$$

Emergence of a Systemic Axis

This result implies:

> A structurally preferred direction (Systemic Axis) must exist within the manifested universe.

The corresponding scalar displacement magnitude is:

$$|\vec{\beta}| = \frac{\Delta H}{H_0} \approx 0.0865 \tag{40}$$

3. From Spatial Displacement to a Temporal Unit

The result above appears, at first glance, as a purely spatial displacement ratio. However, within the SIC / PMM geometric framework, such a displacement is not a simple spatial offset, but rather reflects the **settlement rhythm** of the information manifestation process.

The key principle is:

> **All geometric compensation must ultimately be settled through information propagation delay (τ).**

Therefore, any stable and persistent displacement in the manifested structure must necessarily correspond to a characteristic minimal unit of delay.

In other words:

- $\vec{\beta}$ (spatial compensation)
  $\rightarrow$ represents the minimal displacement requirement within the interlayer
- This displacement, during manifestation
  $\rightarrow$ must be paid as a **delay cost**

Thus, within the PMM framework, this quantity is no longer merely a spatial parameter, but becomes the unique discrete temporal scale required to maintain settlement consistency:

> **Temporal Displacement Unit**

$$\Delta t_{\text{base}} = \beta \tag{41}$$

4. Geometric Interpretation: β as the Minimal Settlement Unit

The physical implication of this transformation is:

- β is no longer just a spatial proportional constant
- It becomes the **minimal time step of all information settlement processes**

Therefore:

- Spatial displacement compensation ($\vec{\beta}$)
- Temporal delay settlement (Δt)

are geometrically dual representations of the same underlying structure.

> β **serves as the bridge between spatial compensation and temporal discretization.**

5. Observable Predictions: Anisotropic Residuals

If β is a real geometric parameter of the system, it must lead to observable consequences:

- The universe is not perfectly isotropic
- There exists a weak but systematic directional bias

This may manifest as:

- Directional dependence in the measured Hubble parameter
- Anomalies in the Cosmic Microwave Background (CMB), such as the Cold Spot
- Asymmetries in large-scale galaxy distributions

> **If multiple independent observations exhibit a consistent preferred direction, this may serve as empirical evidence for the existence of a Systemic Axis.**

6. Derivation of the Information Saturation Limit Σ and Fundamental Length Scale

Introduction:
Based on the entropy saturation principle at the black hole horizon (normalized to unity), this study examines the balance between geometric rigidity and information capacity in local observable space under a non-zero displacement loss β.

Empirical Observation:
A numerical value $\beta \approx 0.0865$ consistently appears across multiple scales, including cosmological deviations, structure formation, and microscopic dynamical phenomena. This recurrence suggests that β is not tied to a specific physical mechanism, but instead reflects a deeper geometric constraint.

Theoretical Position:
In this framework, β is not treated as a fitting parameter. Rather, it is interpreted as an intrinsic loss term arising from geometric mapping, with its value emerging from spatial discretization and symmetry-breaking constraints.

Local Information Capacity Limit ($\Sigma = 11.56$)

If the horizon is defined as the saturation point of information manifestation (normalized to 1), and the observed displacement loss is $\beta = 0.0865$, then to maintain global dynamic consistency, the total information capacity of a single discrete spatial unit Σ must satisfy the following convergence condition:

$$\Sigma = \frac{1}{1-\beta} \times \mathcal{C}_{\text{geom}} \approx 11.56 \tag{42}$$

where $\mathcal{C}_{\text{geom}}$ represents a structural correction factor determined by local geometric configurations (e.g., node connectivity and symmetry constraints), typically of order $\mathcal{O}(1)$.

> **Corresponding fundamental spatial resolution:** $L = \sqrt{\Sigma} = 3.4$

THEORETICAL PREDICTIONS AND FALSIFIABILITY

Based on the displacement vector derived from boundary back-propagation, SIC proposes a set of strongly exclusive predictions. These reflect a conceptual shift: from assuming perfect symmetry to accepting observable displacement as a fundamental feature, and reconstructing the physical framework accordingly.

Cross-validation with multiple experimental datasets suggests that correction terms obtained in conventional perturbative approaches may be reinterpreted as intrinsic geometric effects within the SIC framework, rather than independently introduced approximations.

This further implies that the validity of SIC does not rely on any specific unit system or numerical convention, but instead arises from the use of **dimensionless quantities** as the foundational basis of analysis.

Under this premise, relationships among physical quantities and constants must be understood through structural reinterpretation, rather than direct numerical correspondence in decimal representation. In other words, numerical values serve only as representations of deeper geometric and dynamical structures.

As experimental precision improves, the intrinsic displacement loss parameter $\beta = 0.0865$ and its associated capacity scale $\Sigma = 11.56$ increasingly exhibit a non-negligible and systematic role across multiple scales. Incorporating this term leads to the following geometric settlement structure:

β as a Convergent Solution Under Multiple Constraints

In this model, $\beta \approx 0.0865$ is neither introduced as a predefined constant nor obtained from fitting a single phenomenon. Instead, it emerges from the intersection of multiple independent geometric and physical constraints.

These constraints include, but are not limited to:

- spatial geometric topology (e.g., close-packing structures)
- macroscopic observational residuals (e.g., cosmological-scale discrepancies)
- consistency conditions between local information capacity and transmission cost
- stability requirements of microscopic interaction strengths (e.g., the fine-structure constant)

In general, each constraint allows a finite range of admissible values for β. However, when all constraints are imposed simultaneously:

> **the solution space collapses from a continuous domain into a narrow region, converging toward a single stable point.**

At the current level of modeling precision, this convergent value is:

$$\beta \approx 0.0865 \tag{43}$$

High-Dimensional Uniqueness and Consistency Constraints

In the above derivation, the parameter β is not obtained by fitting any single physical constant. Instead, it emerges as a **unique solution under global geometric constraints**.

The essential implication is:

> **β must simultaneously preserve consistency across multiple physical structures spanning different scales, rather than optimizing any local condition.**

This leads to a critical consequence:

- If β deviates from this unique value, inconsistencies will arise across different physical frameworks, including but not limited to:
  - geometric delay structures in relativity (e.g., time dilation ratios)
  - coupling constants and energy level structures in quantum mechanics
  - expansion rates and observational residuals at cosmological scales

In other words:

> **Any local perturbation of β necessarily induces a breakdown in at least one established theoretical framework.**

Therefore, $\beta \approx 0.0865$ is not a phenomenological fit, but rather the result of a **consistency-locking mechanism** under higher-dimensional constraints.

This property further implies:

- SIC does not rely on introducing additional free parameters to match observations
- Instead, it enforces agreement by **reducing degrees of freedom**, constraining all theories onto a common geometric solution

> **As a result, β becomes a strictly falsifiable parameter: if high-precision experiments at any scale reveal systematic deviations, the entire framework must be revised or rejected.**

Conclusion

> **The value of β is not determined by any single theory, but emerges from the convergence of multiple physical and geometric consistency conditions.**

Its uniqueness, therefore, is not assumed, but arises as a consequence of structural self-consistency.

Global Invariance and Local Effective Deformation of β

Within the SIC / PMM framework, the systemic displacement parameter β is defined as a global geometric quantity that enforces consistency across multiple physical structures. Accordingly, in an ideal homogeneous interlayer:

$$\beta = \text{const.} \approx 0.0865 \tag{44}$$

This value corresponds to a **stable fixed point** of the underlying geometric and informational structure, and cannot be arbitrarily adjusted.

1. Global Invariance

Under large-scale and homogeneous conditions, β must remain fixed in order to ensure:

- stability of microscopic interactions (e.g., the fine-structure constant α)
- consistency of macroscopic geometry (e.g., spacetime expansion structure)
- self-consistency between information capacity (Σ) and delay structure (τ)

> **This invariance operates at the structural level, rather than as a purely numerical constancy.**

2. Local Effective Deformation

Under extreme conditions, the system may exhibit a **local effective variation of** β, which does not correspond to a change in the fundamental constant itself, but instead arises from:

> **local restructuring driven by information density, geometric tension, and expansion rate (Φ).**

In this regime, one must distinguish between the global parameter and its effective local manifestation:

$$\beta_{\text{eff}} = \beta + \delta\beta(\Xi, \nabla\Phi, \omega) \tag{45}$$

where:

- $\delta\beta$ represents an emergent correction induced by local information compression and geometric constraints
- it does not modify the global value of β, but only its effective projection at the observational level

3. Physical Realizations

Such local effective deformation may correspond to several known or anticipated physical regimes:

- **High-density matter (e.g., neutron star interiors):**
  $\Xi \uparrow \rightarrow$ geometric compression $\rightarrow \beta_{\text{eff}} \uparrow$
  $\rightarrow$ leading to modified interaction strengths

- **Strong-field regions (strong gravity / strong electromagnetic fields):**
  $\nabla\Phi \uparrow \rightarrow$ distortion of delay structure $\rightarrow$ effective coupling shifts

- **Low-temperature condensed states (e.g., superconducting systems):**
  information enters highly coherent states $\rightarrow$ reduced geometric impedance $\rightarrow \beta_{\text{eff}} \downarrow$
  $\rightarrow$ transmission cost approaches a minimum

4. Consistency Constraint

Despite the presence of local variations $\delta\beta$, the system must satisfy the global constraint:

$$\langle \beta_{\text{eff}} \rangle_{\text{global}} = \beta \tag{46}$$

That is:

> **all local deviations must cancel out in the global average to preserve overall geometric consistency.**

Conclusion

> β **is a global geometric invariant, while locally it may exhibit effective deformation under specific informational and geometric conditions.**

This structure enables SIC to simultaneously possess:

- **Rigidity:** ensuring cross-scale consistency
- **Flexibility:** allowing diverse local physical phenomena

thereby avoiding both over-constrained rigidity and arbitrary parameter fitting.

[Prediction 01] Cosine Anisotropic Correction of the Cosmic Expansion Rate

Logical Revision:
Once the observer is recognized to be in a displaced state characterized by $\beta = 0.0865$, the so-called "Hubble tension" can be reinterpreted not as an evolutionary inconsistency, but as a purely geometric effect.

$$H_{\text{obs}}(\theta) = H_{\text{true}} \cdot (1 + \beta \cos\theta) \tag{47}$$

$$a_{\text{pioneer}} = H_0 \cdot c \cdot \beta \tag{48}$$

1. **Anisotropy:**
   Predicts that large-scale measurements of the Hubble parameter exhibit directional dependence. The maximum value occurs along the systemic axis direction ($\theta = 0°$), while the minimum appears in the opposite direction.

2. **Informational Inertia:**
   Predicts that all spacecraft exiting the solar system experience a constant "informational inertial drag," with magnitude determined by the above relation (corresponding to the Pioneer anomaly).

   [Falsifiability Criterion]
   If observations confirm that deviations in H_0 are independent of sky direction θ, or if Pioneer-like anomalous deceleration disappears after fully accounting for all thermal effects, the displacement-vector model of SIC is falsified.

[Prediction 02] Mass Locking Mechanism and Geometric Collapse at Absolute Zero

Logical Revision:
Mass is redefined as a geometrically "rotationally locked" state of information ($\vec{\omega}$) within the interlayer. It represents an inertial residual arising from compensation of the global refresh rate $\beta = 0.0865$.

1. Mass Locking Equation:

$$m_{\text{static}} = \frac{\Xi}{\beta_{\text{sys}}} \oint_{\partial\sigma} (\vec{\omega}_{\text{info}} \cdot \vec{n}) dA \tag{49}$$

Note: Under a fixed refresh rate $\beta = 0.0865$, the mass m is determined by the coupling between the information rotation vector $\vec{\omega}$ and the spatial gradient $\nabla\Phi$.

2. Cross-Scale Collapse Threshold (Absolute Zero Collapse):

$$\lim_{T \to 0\,\text{K}} |\vec{\omega}_{\text{info}}| < \beta_{\text{sys}} \;\Rightarrow\; \oint_{\partial\Sigma} (\nabla\Phi \cdot \vec{n} + \vec{\beta}_{\text{sys}}) dA \neq 0 \tag{50}$$

Implication:
As thermal motion approaches zero, if the information rotation $\vec{\omega}$ falls below the systemic refresh threshold (0.0865), geometric consistency fails due to the loss of compensating terms, leading to a form of "numerical overflow" and breakdown of mass-locking states.

- **Geometric Genesis:**
  Predicts that at specific resonance frequencies $f = \beta \cdot n$, pure information density Ξ can condense directly into localized mass-locking states ($m > 0$).
- **Structural Dissolution:**
  Predicts that below a critical temperature $T < T_{\text{crit}}$, matter ceases to exhibit particle-like behavior and instead transitions into a "pure geometric wave" with a characteristic 180° symmetry bias.

[Prediction 03] Background-Flow Modulation of Gravitational Wave Propagation

Logical Revision:
Gravitational waves, as ripples of spatial information, propagate within a background that itself possesses a displacement flow. Their phase and velocity must therefore be modulated by this background bias. This is not a conventional Doppler effect, but a direct consequence of geometric flow influencing the waveguide structure.

1. Modulation Equations:

$$v_{\text{gw}}(\theta) = c \cdot (1 + \beta \cos\theta) \tag{51}$$

$$\Delta\phi_{\text{gw}} = \int_L \vec{\beta} \cdot d\vec{l} \tag{52}$$

1. **Propagation Anisotropy:**
   Predicts an $\mathcal{O}(\beta)$-level directional dependence in gravitational wave propagation speed across different sky regions.
2. **Phase Drift:**
   Predicts that for gravitational wave events traversing cosmological distances, waveform reconstruction must include cumulative phase corrections induced by the displacement vector $\vec{\beta}$. Otherwise, inferred source parameters (e.g., mass, distance) will exhibit systematic bias.

   [Falsifiability Criterion]
   If future space-based observatories (e.g., LISA) confirm that gravitational wave propagation speed and phase are fully isotropic, with no systematic dependence on direction or displacement structure, the SIC framework is falsified.

[Prediction 04] Attosecond Displacement Settlement and Electron Localization

Structural Derivation:
After subtracting the displacement compensation induced by the system refresh rate $\beta = 0.0865$, the motion of the electron at the attosecond scale follows a **deterministic geometric trajectory**, rather than a statistical probability distribution.

This marks a reinterpretation of quantum uncertainty:
the so-called wavefunction is merely a **residual projection under high-frequency settlement dynamics**.

1.Electron Trajectory Reconstruction Equation:

$$\vec{r}_{\text{real}}(t) = \vec{r}_{\text{obs}}(t) - \int_{t_0}^{t_1} (c \cdot \beta_{\text{sys}} \cdot \vec{n}_\theta) \, dt \tag{53}$$

Note:
When observational resolution reaches the attosecond regime ($\sim 10^{-18}$ s),
the dynamic blurring induced by $\beta = 0.0865$ vanishes, revealing a continuous trajectory consistent with causality.

[Falsification Criterion]
If, after applying the β correction in attosecond experiments, the electron density distribution remains irreducibly stochastic and does not converge to a single geometric flow line, then the deterministic framework of SIC is invalidated.

[Prediction 05] Geometric Origin of the Anomalous Magnetic Moment and Discrete Mapping of the Fine-Structure Constant α

Summary:
This prediction proposes that the electron anomalous magnetic moment $(g-2)$ and the fine-structure constant α do not originate from infinite-order perturbative expansions, but instead arise from a **discrete geometric settlement structure** at the microscopic resolution scale ($L = 3.4$).

Within the SIC / PMM framework, electron phase evolution does not occur on a continuous manifold, but is constrained by:

- **Displacement baseline loss rate:** $\beta = 0.0865$
- **Information capacity bound:** $\Sigma = 11.56$

These jointly define the minimal unit of geometric settlement and the capacity boundary, implying that any periodic motion (such as spin phase evolution) necessarily accumulates a **non-removable geometric residual upon closure**.

1.Geometric Correction Equation for the Electron Anomalous Magnetic Moment:

$$a_e = \frac{g-2}{2} = \frac{1}{\Sigma} \oint_0^{2\pi} \beta \cdot f(\cos\theta) \, d\theta \tag{54}$$

Geometric Interpretation:

- Electron spin is not merely an SU(2) symmetry operation, but a **closed settlement path on a discrete geometric lattice**
- Each infinitesimal angular displacement incurs a minimal delay cost β
- Upon completing a 2π phase cycle, the accumulated delay cannot fully cancel, yielding the observed a_e

Thus, $(g-2)$ is not purely a quantum fluctuation effect, but a macroscopic projection of "incomplete discrete geometric settlement."

2.Geometric Projection Representation of the Fine-Structure Constant:

$$\alpha \approx \frac{\beta}{4\pi\Sigma} \tag{55}$$

Refined Interpretation:

- α is no longer a fundamental constant, but:

  the effective projection ratio of information completing a closed transmission within a unit geometric capacity

- 4π corresponds to the minimal closed symmetry path in three-dimensional space
- Σ represents the maximum information capacity of a spatial unit
- β represents the minimal delay cost per transmission step

This relation implies:

The strength of electromagnetic coupling is fundamentally a ratio between geometric capacity and settlement cost, rather than the result of virtual particle exchange.

3.Dynamic Corrections and Scope of Validity:

It is essential to emphasize:

- β is a **global geometric base solution**, not continuously tunable
- However, under **extreme local conditions** (high density, strong fields, phase transitions):
  - Effective Σ_{local} decreases
  - Apparent coupling strength increases

This is analogous to:

effective parameter renormalization in superconductivity or strongly correlated systems, rather than a modification of fundamental constants.

[Falsifiability Criterion]

In high-energy scattering or extreme-field experiments:

If observations show:

- Coupling constants fully follow continuous renormalization group (RG) flow
- And no discrete steps or non-smooth transitions appear near extreme scales related to $\Sigma = 11.56$

$\rightarrow$ The model is falsified.

Conversely, if high-precision measurements reveal:

- Non-continuous coupling shifts
- Or scale-specific structural residuals

$\rightarrow$ These would serve as evidence supporting the geometric settlement framework.

[Prediction 06] Cross-Energy Photoelectric Coupling and Underlying Code Coverage

Structural Derivation:

By mapping the experimentally observed photoelectric range (1.5–6.0 eV) into the SIC framework, it is revealed that what is conventionally interpreted as "statistical thermal loss" originates from neglecting a global phase compensation factor $\cos(\beta)$, where $\beta = 0.0865$ represents the angular offset between the observational frame and the underlying cosmological metric.

Once this compensation is applied, conventional physical models naturally reduce to a lower-dimensional subset of the SIC formalism.

> **Energy Mapping:**
> The interval (3.4–11.56 eV) corresponds precisely to the projection of geometric resolution L and information capacity Σ into the energy domain, forming the effective spectral bandwidth of the system.

1.Inverse-Entropy Field Energy Coverage Equation:

$$E_{\text{SIC}} = \int_{3.4}^{11.56} \left[\Phi_{\text{classical}}(\omega) + \Delta S_{\text{loss}}(\beta)\right] d\omega \tag{56}$$

Interpretation:

When the upper energy bound approaches 11.56 eV and the $\beta = 0.0865$ phase compensation is applied, the term ΔS_{loss}—previously interpreted as thermal dissipation—transforms into a coherent energy contribution.

This suggests that the conventional photoelectric formulation represents a statistical approximation under phase-misaligned conditions. With proper phase alignment, information previously distributed across entropy channels can be reallocated into observable degrees of freedom.

> **[Falsification Criterion]**
> If, within the 3.4–11.56 eV bandwidth and under β phase correction, photoelectric conversion remains constrained by the Shockley–Queisser limit and continues to exhibit irreversible entropy increase, without evidence of coherent recovery from thermal loss, then the SIC code-completeness hypothesis is invalid.

[Prediction 07] Black Hole Entropy Mapping and Macro–Micro Spatial Decoupling

Structural Derivation:

Within the SIC framework, black holes are interpreted as macroscopic projections of microscopic states. By inverting black hole entropy observations, the unit displacement loss (phase taxation rate) is identified as $\beta = 0.0865$, leading to a maximal information capacity per spatial unit:

$$\Sigma = \frac{1}{\beta} = 11.56 \tag{57}$$

Since black hole entropy scales with surface area, this defines a macroscopic vacuum length scale:

$$L = \sqrt{11.56} = 3.4 \tag{58}$$

These values (3.4 and 11.56) are interpreted as artifacts of decimal representation rather than fundamental constants.

When the displacement correction term $0.0865c$ is applied to reconstruct microscopic dynamics, the system converges toward the natural base:

$$e \approx 2.718... \tag{59}$$

1.Macro–Micro Separation and Entropy Capacity Equations:

$$\Sigma_{\text{capacity}} = \frac{1}{\beta} = 11.56 \quad \text{(decimal representation: surface entropy limit)} \tag{60}$$

$$L_{\text{macro}} = \sqrt{\Sigma_{\text{capacity}}} = 3.4 \quad \text{(decimal representation: vacuum metric)} \tag{61}$$

$$L_{\text{micro}} = f(0.0865c) \to e \quad \text{(natural representation: microscopic basis)} \tag{62}$$

Interpretation:
This prediction establishes the necessity of "decimal decomposition." When the displacement correction $0.0865c$ is incorporated, macroscopic constants collapse into pure ratios defined on an exponential (e-based) lattice.

This process separates:

- **Vacuum background (geometric substrate)**
- **Matter rendering (projected structure)**

into distinct computational layers.

[Falsification Criterion]
If future observations show that black hole entropy does not exhibit a reciprocal structure consistent with $1/0.0865$, or if, after removing decimal artifacts, the minimal unit of microscopic displacement does not converge toward e, then the macro–micro separation mechanism is invalid, and the SIC framework fails at the foundational level.

CORE DERIVATION: GEOMETRIC ORIGIN AND DYNAMIC COUPLING OF THE FINE-STRUCTURE CONSTANT α

(Geometric Origin and Dynamic Coupling of the Fine-Structure Constant α)

Core Perspective

In standard physics, the fine-structure constant $\alpha \approx 1/137$ is treated as a dimensionless parameter that cannot be derived from within the theory itself, serving as a measure of electromagnetic interaction strength.

Within the SIC framework, however, α is no longer regarded as an empirical input, but is reinterpreted as:

The coupling projection rate of information as it propagates through available geometric channels within the spatial interlayer under specific structural constraints.

In this sense, α is not a fundamental constant, but an emergent quantity determined jointly by geometric structure and information capacity.

1. Geometric Base and Residual Extraction

(Geometric Base and Residual Extraction)
This derivation is built upon two independent yet cross-consistent geometric inputs:

- **System displacement base β:**
  A dimensionless parameter extracted from macroscopic observational residuals (e.g., Hubble tension):

$$\beta \approx 0.0865 \tag{63}$$

- **Geometric channel number (Kissing Number):**
  In three-dimensional close-packing geometry, the maximum number of stable nearest neighbors is:

$$N_{\text{channel}} = 12 \tag{64}$$

This value represents the **maximum number of independent geometric channels** available for stable information transmission within the local interlayer.

2. Capacity–Cost Duality

(Capacity–Cost Duality)
Within the SIC projection framework, information capacity (Σ) and displacement delay cost (β) satisfy a geometric duality:

$$\Sigma = \frac{1}{\beta} \tag{65}$$

Substituting $\beta = 0.0865$:

$$\Sigma \approx 11.56 \tag{66}$$

Physical Interpretation:

- Σ represents the maximum information capacity per spatial unit (manifestation capacity)
- β represents the minimal delay cost required for a unit geometric displacement

This relation implies:

As displacement becomes more costly ($\beta \uparrow$), available capacity decreases ($\Sigma \downarrow$).

3. Coupling Constant as Channel Projection

(Coupling as Channel Projection)
Within this framework, electromagnetic interaction strength does not arise from a single pathway, but rather from:

The total projection efficiency of information propagating through all available geometric channels.

Thus, the inverse fine-structure constant can be expressed as:

$$\alpha^{-1} = N_{\text{channel}} \cdot \Sigma = 12 \cdot \frac{1}{\beta} \tag{67}$$

Substituting values:

$$\alpha^{-1} \approx 12 \times 11.56 \approx 138.7 \tag{68}$$

Accounting for local geometric perturbations and inhomogeneities, this value converges toward the experimental result:

$$\alpha^{-1} \approx 137 \tag{69}$$

4. Three-Regime Dynamic Coupling

(Three-Regime Dynamic Coupling)
Since α depends on both β and Σ, it is fundamentally a **state-dependent quantity**, rather than an absolute constant.
Depending on information density and spatial compression, the system exhibits three characteristic regimes:

Normal Vacuum Regime

- $\beta \approx 0.0865$
- $\Sigma \approx 11.56$
- $\alpha^{-1} \approx 137$

Interpretation:
Geometric channels are fully accessible, information transfer cost is low, and electromagnetic interaction assumes its standard strength, allowing stable atomic structures.

High-Density Compressed Regime (e.g., neutron star interiors)

- $\beta \uparrow\sim 0.15 - 0.2$
- $\Sigma \downarrow\sim 5 - 6$
- $\alpha^{-1} \sim 60$

Interpretation:
Geometric channels are compressed, increasing transmission cost and enhancing effective coupling strength.
The distinction between electromagnetic and strong interactions begins to blur, and stable atomic structures can no longer be sustained.

Critical Meltdown Regime ($\beta \to 1$)

- $\beta = 1$
- $\Sigma = 1$
- Effective channel number $\to 1$
- $\alpha^{-1} \to 1$

Interpretation:

- All geometric channels collapse into a single degree of freedom
- All interactions become geometrically equivalent
- The system enters a limit where manifestation fails (logical breakdown state)

β and the Residual Structure of α

Within the present framework, the base temporal displacement parameter β is constrained through multiple independent experimental signatures, including interference structures, temporal coherence, and manifestation frequency. Its numerical stability across different physical contexts suggests that it plays the role of a fundamental structural parameter.

In contrast, while the fine-structure constant α is experimentally determined with extremely high precision, its geometric reconstruction within the current model exhibits a small but systematic deviation. This discrepancy should not be interpreted as numerical inaccuracy, but rather as an indication that:

α may encode higher-dimensional phase-structural corrections not yet explicitly resolved within the present formulation.

From the PMM / SIC perspective:

- β corresponds to the base unit of temporal manifestation
- α may represent an effective coupling constant emerging from higher-dimensional phase projections

For this reason, we deliberately refrain from forcing an exact derivation of α. Preserving this gap maintains consistency with empirical data while leaving room for the incorporation of additional phase degrees of freedom, particularly those associated with temporal structure beyond the current dimensional resolution.

Conclusion

> **The fine-structure constant α is not an arbitrary fundamental constant, but a projection outcome determined by the geometric channel number (12) and the spatial information capacity (Σ). What is commonly interpreted as force unification is not a change in interaction laws, but a geometric degeneration of channel structure under extreme information density, causing all interactions to converge onto a single projection scale.**

Supplementary Note: Consistency Between $\Sigma = \frac{1}{\beta}$ and $\Sigma = \frac{1}{1-\beta}$

In the preceding derivations, the information capacity Σ appears in two distinct forms:

$$\Sigma = \frac{1}{\beta} \quad \text{and} \quad \Sigma = \frac{1}{1-\beta} \tag{70}$$

This difference does not represent a contradiction, but rather reflects two complementary geometric perspectives.

1. Transport Perspective

When Σ is defined as the inverse of the geometric cost per information transfer, the focus is on:

The delay cost required for information to traverse a single fundamental unit.

In this case:

$$\Sigma = \frac{1}{\beta} \tag{71}$$

Interpretation:

- β: minimal delay cost per elementary displacement
- Σ: maximum number of transmissible steps (or effective capacity) under a fixed total budget

2. Capacity / Saturation Perspective

When Σ is defined as the total information capacity before reaching the manifestation limit, the focus shifts to:

The remaining margin before geometric saturation ($\beta \to 1$).

In this case:

$$\Sigma = \frac{1}{1-\beta} \tag{72}$$

Interpretation:

- $1 - \beta$: remaining margin before geometric saturation or collapse
- Σ: total accumulable capacity prior to instability

3. Unified Geometric Interpretation

These two formulations describe complementary aspects of the same underlying structure:

Perspective	**Physical Meaning**	**Mathematical Form**
Transport Cost	Cost per step	$\Sigma = 1/\beta$
Remaining Capacity	Available margin	$\Sigma = 1/(1-\beta)$

This reflects a fundamental duality:

The complementarity between local transmission cost and global capacity margin.

4. Consistency Condition

To avoid ambiguity, the SIC framework imposes the following constraint:

> **In any specific derivation, the physical role of Σ must be explicitly specified (transport or capacity), and the two definitions must not be mixed within the same logical step.**

In the derivation of the fine-structure constant α presented in this work:

- The **transport perspective** is adopted
- Therefore:

$$\Sigma = \frac{1}{\beta} \tag{73}$$

Conclusion

> $\Sigma = 1/\beta$ **and** $\Sigma = 1/(1-\beta)$ **are not competing definitions, but dual representations of the same geometric structure under different observational regimes.**

The former describes **how information propagates**, while the latter describes **how much the system can sustain before saturation**.

[Foundational Protocol] Dimensionless Structure in SIC and Convergence to the Natural Base

Core Logic:
All structural scales in SIC originate from a single dimensionless parameter, $\beta = 0.0865$. This defines a closed and self-consistent hierarchical system that governs the full process from microscopic information processing to macroscopic geometric manifestation.

In the native representation of the universe, numerical values are not mere counting units, but **ratio factors associated with spatial folding and phase compensation**.

1. Fundamental Definition (Dimensionless Basis)

$$\beta := 0.0865 \tag{74}$$

Definition: Displacement loss rate (phase tax).
It acts as the fundamental source of observable deviations in physical measurements (e.g., Hubble tension, anomalous magnetic moments).

2. Capacity (Structural Upper Bound)

$$\Sigma \equiv \beta^{-1} \approx 11.56 \tag{75}$$

Definition: Dimensionless capacity; the saturation limit of information storage within a spatial unit.
At the macroscopic level, this corresponds to the entropy–area bound observed in black hole systems.

3. Macroscopic Scale (Geometric Resolution)

$$L \equiv \Sigma^{1/2} = \beta^{-1/2} \approx 3.4 \tag{76}$$

Definition: Dimensionless characteristic length; the geometric resolution scale.
It defines the minimal effective granularity of the vacuum background accessible to observation.

4. Microscopic Generative Base (Phase–Manifestation Mapping)

When the displacement correction term $0.0865c$ is introduced to reconstruct microscopic trajectories, the system naturally converges from decimal representation to the intrinsic base e:

$$e \equiv \lim_{\beta \to \beta_0} \Phi(\beta) \tag{77}$$

$$L_{\text{micro}} = f(0.0865c) \to e \approx 2.718... \tag{78}$$

Mathematical Convergence Property:

$$e \equiv \exp(1) = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n \quad \longleftrightarrow \quad \text{phase accumulation under normalized structure} \tag{79}$$

Summary:

The commonly used values 3.4 and 11.56 are merely **transitional representations induced by decimal conventions**.

Once the phase offset $\beta = 0.0865$ is accounted for, macroscopic geometric quantities collapse into pure ratios defined over the natural base e.

This reveals a fundamental separation between:

- **Vacuum background (geometric substrate)**
- **Matter rendering (manifested structure)**

at the algorithmic level.

[ULTIMATE IMPLICATION] INCOHERENT SEPARATION BETWEEN COSMOLOGICAL DISPLACEMENT AND MICROSCOPIC REPRESENTATION

Structural Derivation

A key turning point in this framework is that SIC does not rely on specific unit systems or numerical choices, but is instead constructed entirely upon **dimensionless quantities**. Under this condition, all physical quantities and constants must be structurally reinterpreted according to their respective hierarchical origins, rather than directly compared within a decimal numerical system.This transformation allows mixed quantities at the representation level to be decomposed back into their generative sources.

Geometric Role of Background Displacement

Within this framework, the cosmological displacement:

$$\beta = 0.0865 \tag{80}$$

is no longer interpreted as a simple velocity ratio, but as a geometric parameter describing **systematic deviation in signal propagation relative to the reference speed** c. At the observational level, its effect can be equivalently described as propagation delay or refractive behavior in an effective medium. However, such refractive interpretation is merely phenomenological, not fundamental.

Incoherent Dual-Channel Structure

The system separates into two independent channels:

- **Macroscopic Channel**
  Governed by β, determining structural quantities such as Σ and L
- **Microscopic Channel**
  Governed by intrinsic dynamical rules, with manifestation expressed through phase evolution (e.g., $e^{i\theta}$)

These two channels are **non-coherent**, meaning:

> Their generative mechanisms, parameter origins, and settlement rules are fundamentally independent and cannot be directly mapped within a single representation space.

Origin of Representation Mismatch

Human reliance on decimal representation leads to direct numerical comparison and superposition of quantities from different structural layers, for example:

- Macroscopic scale: $L \approx 3.4$
- Microscopic base: $e \approx 2.718$

Such operations are formally valid but structurally inconsistent, as they mix quantities from different generative levels.

Once reformulated in a dimensionless structural space, these quantities naturally separate and converge to their respective origins, eliminating artificial numerical coupling.

Conclusion

This analysis shows that certain physical constants that appear related are in fact generated from **distinct structural layers**.

Their numerical proximity does not imply fundamental unity, but rather reflects projection effects induced by the representation system.

Through dimensionless formulation and structural decomposition, these mixed phenomena can be separated into independent channels, restoring their proper physical interpretation.

PHASE-MANIFESTATION MECHANICS: FROM OBSERVATIONAL CORRECTION TO OPERATOR FORMALISM (FINAL SYNTHESIS)

Phase-Manifestation Mechanics (PMM)

This section synthesizes the preceding experimental observations and theoretical refinements. Under the conditions of dimensionless formulation and structural consistency, the framework is reformulated within SIC (Spatial Interlayer Correlation).

Physical phenomena are no longer treated as outputs of a single causal chain, but rather as **manifestations arising under specific geometric and phase conditions**.

CORE FRAMEWORK TRANSITION: FROM "QUANTITATIVE EQUATIONS" TO "MANIFESTATION OPERATORS"

Theoretical Revision

Conventional physics typically describes observations in the form of *numerical values multiplied by dynamical terms*.
Within the SIC framework, however, physical systems are more appropriately interpreted as **generative structures**:

> What is commonly referred to as "matter" is not a fundamental entity undergoing independent motion, but a stable manifestation formed under specific conditions between a background phase flow and a structural substrate.

Structural Reformulation

By introducing dimensionless variables:

$$\tilde{E} = \frac{E}{E_{\Sigma}}, \quad \tilde{r} = \frac{r}{L} \tag{81}$$

all observables can be reinterpreted as **geometric projections in intrinsic structural coordinates**.
Under this perspective:

> Reality is no longer an isolated numerical outcome, but a confirmed manifestation emerging from the interplay between global structure and local phase.

$$\text{Reality} = \mathcal{P}_{\beta}\left(e^{i\theta(\text{background})},\, \Sigma, L\right) \tag{82}$$

Key Parameter Definitions

- **Σ (Capacity) and L (Resolution):**
  Define the structural limits and spatial resolution of the system, forming the geometric substrate of manifestation.
- **$e^{i\theta(\textbf{background})}$ (Background Phase Flow):**
  θ encodes the dynamical evolution of the background, representing phase variation external to the structural layer.
- **$\mathcal{P}_{\beta}$ (Manifestation Operator):**
  β acts as the control parameter governing coupling and manifestation conditions, determining how phase information is mapped into observable structures.

CONCLUSION: EXISTENCE AS A PROJECTION CONDITION

Structural Conclusion

Within this framework, physical reality is no longer interpreted as the direct result of object motion, but as a **stable manifestation arising when background phase flow interacts with a structural substrate under specific conditions**.

This shifts the focus of physical description:

- from **trajectories and forces**
- to **phase structures and manifestation conditions**

When a mismatch exists between background phase and structural substrate, the resulting phase difference manifests observationally as:

- energy distributions
- mass properties
- temporal scales

This formulation does not seek to replace existing theories, but rather to provide a **unified projection-based interpretation under a dimensionless structural framework**.

CORE STRUCTURAL TRANSITION: FROM QUANTITATIVE EQUATIONS TO MANIFESTATION OPERATORS

Theoretical Revision

In conventional physics, observed phenomena are typically expressed in the form of *numerical value* × *dynamical term*. However, within the SIC framework, physical systems are more appropriately understood as a **generative structure**:

> What is referred to as "matter" is not an independently evolving entity, but a **stable manifestation** arising from the interaction between background phase flow and structural substrate under specific conditions.

Structural Reformulation

By rendering all physical quantities dimensionless:

$$\tilde{E} = \frac{E}{E_\Sigma}, \quad \tilde{r} = \frac{r}{L} \tag{83}$$

all observables can be reinterpreted as **geometric projections within intrinsic structural coordinates**.
In this representation:

- Reality is no longer an isolated numerical output
- but a **manifested state** jointly determined by global structure and local phase dynamics

$$\text{Reality} = \mathcal{P}_\beta\left(e^{i\theta(\text{background})}; \Sigma, L\right) \tag{84}$$

Key Parameter Definitions

- Σ **(Capacity)** and L **(Resolution)**
  Define the structural bounds and spatial resolution scale of the system, forming the geometric substrate of manifestation.
- $e^{i\theta(\textbf{background})}$ **(Background Phase Flow)**
  θ encodes the dynamical evolution of the background, representing phase variation external to the structural layer.
- $\mathcal{P}_\beta$ **(Manifestation Operator)**
  β acts as the governing parameter for coupling and projection, determining how phase information is mapped into observable structure.

CONCLUSION: EXISTENCE AS A PROJECTION CONDITION

Structural Conclusion

Within this framework, physical reality is no longer interpreted as the direct outcome of "objects in motion", but as:

> **a stable manifestation emerging when background phase flow interacts with a structural substrate under specific conditions.**

This shifts the focus of inquiry:

- from **trajectories and forces**
- to **phase structures and manifestation conditions**

When a mismatch exists between background phase and structural substrate, the resulting phase difference manifests observationally as:

- energy distributions
- mass expressions
- temporal scales

This formulation does not replace existing theories, but instead provides a **unified projection perspective under a dimensionless structural framework**.

THE STRUCTURAL ORIGIN OF NEWTONIAN MECHANICS: FORCE AS A DESCRIPTIVE PROJECTION

Historical Position

The classical mechanics established by Isaac Newton provided the first mathematically precise framework capable of predicting the motion of physical objects. By introducing a correspondence between motion and "force," physical phenomena became systematically describable and computable.

Its core formulation can be expressed as:

$$F = ma \tag{85}$$

This relation successfully translates dynamical behavior into measurable quantitative terms, achieving high predictive accuracy at macroscopic scales.

Structural Limitation

However, within this framework, "force" is treated as the direct cause of motion, yet its own generative mechanism remains undefined. In other words, classical mechanics provides rules for describing dynamics, but does not address the origin of those rules.

This leads to a structural characteristic:

- The behavior of physical systems can be accurately predicted
- Yet the question of **why such behavior arises** remains at the level of externally imposed parameters

Reinterpretation under the SIC Framework

Within the SIC framework, classical mechanics can be reinterpreted as:

> A representation method that encapsulates manifested outcomes (observed motion) using effective parameters (force)

That is:

- "Force" is not a fundamental generator of dynamics
- Rather, it is a projected result arising from the interaction between background structure and manifestation conditions

Thus, what Newtonian mechanics captures is not the generative mechanism of the system, but the stable behavioral pattern that emerges under specific manifestation conditions.

Transition

This framework—describing outcomes via phenomenological parameters—was gradually revised in subsequent developments.

The concept of fields, introduced by Michael Faraday, began to reinterpret "force" as a structural distribution within space itself, rather than an externally applied quantity acting between objects.

This shift marks the first step from a **particle-force paradigm** toward a **spatial-structure paradigm** in physical theory.

FARADAY: FROM "FORCE" TO "SPATIAL STRUCTURE"

Faraday's contribution was not limited to the discovery of electromagnetic phenomena; rather, it marked a fundamental conceptual shift in physics:
transforming "force" from a direct interaction between objects into a **structural distribution of space itself**.

In his concept of **field lines**, electricity and magnetism are no longer treated as forces between particles, but as **continuous and spatially distributed geometric structures**.
The motion and interaction of objects are merely manifestations of this structure under specific conditions.

Within the SIC framework, this viewpoint can be further interpreted as:

> Physical phenomena do not arise from interactions between objects, but from the misalignment between the **background structure (Base)** and the **phase flow (Phase Flow)**, producing observable projection effects.

In other words, Faraday implicitly revealed that:

> Space is not a passive container, but an active structural field that generates phenomena.

This transformation laid the foundation for later developments such as electromagnetic wave theory (Maxwell) and modern field theory, and directly aligns with the SIC principle that:

> **Existence is a manifestation outcome.**

Further Interpretation of Phase Structure: Geometric Correspondence of Charge and Current

After Faraday reinterpreted "force" as a distribution of fields, electromagnetic phenomena can be viewed as expressions of spatial structure.
On this basis, SIC provides a deeper geometric interpretation by mapping sources and flows to local features of phase structure.

1. Charge (Source of Phase Curvature)

$$\nabla^2\theta \sim \frac{\rho}{\Sigma \cdot L} \tag{86}$$

Here, θ represents the background phase, and ρ denotes the corresponding source density.
This relation plays the role of a "source term" in field equations, indicating that **charge can be understood as a localized curvature or concentration of phase structure**.

2. Current (Phase Geometric Flow)

$$\mathbf{J}_{\text{phase}} \sim (\partial_t\theta), \nabla\theta \tag{87}$$

Current is no longer interpreted as particle motion, but as the **directed evolution of phase structure in space**, resulting from the coupling between temporal variation and spatial gradient of the phase.

Structural Interpretation

- **Σ (Capacity Constraint):**
  Determines the maximum local curvature that phase structure can sustain, corresponding to the upper bound of stable manifestation.
- **Phase Gradient $\nabla\theta$:**
  Represents the geometric origin of field strength; its direction and magnitude determine observed effects.
- **Temporal Evolution $\partial_t\theta$:**
  Governs the dynamical aspect of the structure, corresponding to the behavior of current.

This description does not replace classical electromagnetic equations, but instead provides a **geometric-layer correspondence**, allowing both "source" and "flow" to be understood as different manifestations of phase structure under varying conditions.

GEOMETRIC REFORMULATION OF MAXWELLIAN DYNAMICS: SELF-CONSISTENT PHASE OSCILLATION

Conceptual Extension

Maxwell's equations successfully describe the coupling and propagation of the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$, and predict the existence of electromagnetic waves. However, their formulation is primarily based on relationships between field quantities, without explicitly defining the underlying structural origin of these fields.

Within the SIC framework, $\mathbf{E}$ and $\mathbf{B}$ can be reinterpreted as projections of a single background phase field θ along different geometric directions:

- $\mathbf{E} \sim \partial_t \theta$ (temporal variation of phase)
- $\mathbf{B} \sim \nabla \theta$ (spatial gradient of phase)

This correspondence unifies the electromagnetic field as a geometric manifestation of a single scalar phase field θ.

1. Phase Dynamics (Corresponding to Source-Free Maxwell Equations)

$$\partial_t^2 \theta - c^2 \nabla^2 \theta = 0 \tag{88}$$

This equation corresponds to the wave equation for electromagnetic waves in vacuum, indicating that in the absence of sources, the background phase can form self-consistent, propagating wave solutions.

2. Manifestation Mapping (Observation Layer)

$$\text{Reality} = \mathcal{P}_\beta\big(e^{i\theta}; \Sigma, L\big) \tag{89}$$

This mapping describes how phase structures, under the control of β, are projected into observable electromagnetic phenomena.

Structural Interpretation

- **Self-Consistency:**
  $\partial_t(\nabla\theta) = \nabla(\partial_t\theta)$
  This expresses the commutativity between temporal evolution and spatial structure, corresponding to the mutual generation of electric and magnetic fields.
- **Role of the Speed of Light c:**
  In this framework, c represents the propagation ratio of phase disturbances within the structural substrate. Its value emerges from the interplay between spatial resolution scale L and phase response time.

Theoretical Interpretation

Electromagnetic waves can be understood as self-sustaining modes of the background phase structure, rather than independent carriers of energy. Their interaction with matter arises from the competition and coupling between different phase solutions under shared manifestation constraints.

In other words:

> Light and matter are not fundamentally distinct entities, but rather different solutions of the same phase-generation mechanism under different boundary conditions.

GEOMETRIC REFORMULATION OF EINSTEINIAN GRAVITY: PHASE COMPRESSION AND RECONFIGURATION OF THE BACKGROUND

Conceptual Extension

In standard general relativity, gravity is described as the geometric curvature of spacetime induced by mass-energy:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{90}$$

However, this formulation remains at the level of observed geometry, without specifying the underlying generative mechanism of curvature.

Within the SIC framework, gravity is no longer interpreted as spacetime curvature, but instead as:

> **The responsive compression of the background phase θ under finite capacity Σ and resolution L, in the presence of high information density.**

Core Reconstruction

- **Mass / Energy:**
  Defined as the localized concentration of information overlap density Ξ
- **Spatial Curvature:**
  Not a priori geometry, but a **nonlinear phase compression** of θ under structural constraints
- **Gravitational Effect:**
  Emerges from gradients induced by phase compression, not from an independent force

1. Phase Field Equation (Correspondence to Einstein Field Equations)

$$\mathcal{R}(\theta) = \kappa \cdot \frac{\Xi}{\Sigma \cdot L} \tag{91}$$

Parameter Definitions:

- $\mathcal{R}(\theta)$: Local curvature (compression) of the background phase
- Ξ: Information overlap density
- $\Sigma \cdot L$: Geometric constraint (capacity × resolution)

This equation describes:

> **How information density is transformed into deformation of phase structure under system constraints.**

As Ξ approaches its limit (e.g., black holes), resolution L breaks down, and manifestation ceases.

2. Manifestation Mapping (Observation Layer)

$$\text{Reality} = \mathcal{P}_\beta\left(e^{i\theta(\Xi)}; \Sigma, L\right) \tag{92}$$

What is observed as a "gravitational field" is fundamentally:

The manifested projection of a compressed phase structure under the operator $\mathcal{P}_\beta$

Structural Interpretation

- **Coupling Coefficient κ:**
  Represents the conversion strength between information density and phase compression
  (geometric response of the interlayer)
- **Equivalence Principle (Reinterpreted):**
  Acceleration and gravity are equivalent because both modify
- **Time Dilation:**
  In regions of strong phase compression, the evolution rate $\partial_t \theta$ decreases
  $\rightarrow$ longer manifestation cycles
  $\rightarrow$ observed as slower time

Theoretical Closure

Gravity is not a fundamental force, but:

A structural response of the background phase under high information density to maintain manifestation consistency

Objects do not follow geodesics because spacetime is curved, but because:

Phase flow selects paths of minimal geometric resistance for stable manifestation

Unified Perspective

Through this reformulation:

- Gravity (macroscopic)
- Quantum coherence (microscopic)

are no longer separate domains, but:

Different manifestations of the same background phase mechanism under different resolutions L

GEOMETRIC REFORMULATION OF SCHRÖDINGER DYNAMICS: PHASE EVOLUTION AND MANIFESTATION PROJECTION

Conceptual Extension

In conventional quantum mechanics, the wavefunction ψ is interpreted as a probability amplitude describing the state of a particle, with its evolution governed by the Schrödinger equation. However, the process of "measurement," which leads to wavefunction collapse, lacks an intrinsic and continuous geometric mechanism.

Within the SIC framework, ψ is no longer interpreted probabilistically, but is instead reduced to an expression of the underlying background phase θ:

$$\psi = e^{i\theta} \tag{93}$$

In other words, physical systems are not "particles moving in space," but rather:

> **The evolution of a background phase under structural constraints (Σ, L), which is projected into reality under specific conditions.**

- **Phase as Generative Structure:**
  $\psi = e^{i\theta}$ represents a continuously evolving phase structure, not a probability distribution.

- **Measurement as Projection:**
  "Observation" is not an external intervention, but the process by which the phase structure is forced into an observable state when the coupling condition β is satisfied.

1. Phase Dynamics Equation (Underlying Evolution)

$$\frac{\partial \theta}{\partial \tilde{t}} = \mathcal{F}(\Sigma, L, \nabla^2\theta, \tilde{V}) \tag{94}$$

Parameter Definitions

- θ: Underlying background phase (not directly observable)
- $\mathcal{F}$: Phase evolution rule (replacing the traditional Hamiltonian structure)
- Σ, L: System capacity and spatial resolution (geometric constraints)
- $\tilde{V}$: Dimensionless potential term

This equation describes:

> **The actual computational process occurring before anything becomes observable.**

2. Manifestation Mapping (Observed Reality)

$$\text{Reality} = \mathcal{P}_\beta\left(e^{i\theta(\tilde{x},\tilde{t})}; \Sigma, L\right) \tag{95}$$

This mapping implies:

> **Reality does not exist a priori, but emerges as the output of phase structure once conditions are satisfied.**

- **From "Probability" to "Manifestation"**
  In this framework, probabilistic interpretation is no longer required.
  The system has only two states:
  - Conditions not satisfied $\rightarrow$ unmanifested (appears as superposition)
  - Conditions satisfied $\rightarrow$ projected into reality via $\mathcal{P}_\beta$
- **Dimensionless Reformulation**
  Through Σ and L, all quantum behavior is unified as a structural balance problem under
  (system capacity × spatial resolution)

Theoretical Interpretation

Quantum mechanics is thus reinterpreted as:

> **Not a theory of how particles move, but a theory of when reality manifests.**

The traditional Schrödinger equation describes only the evolution of phase within structure;
what we observe as physical phenomena is the manifested projection of this evolution under geometric constraints.

Fundamental Shift

The most critical transformation is:

> **"Measurement" is no longer an external disturbance, but an inevitable geometric transition intrinsic to the system.**

STRUCTURAL INTERPRETATION OF THE DIRAC EQUATION: INTERNAL DEGREES OF PHASE

Conceptual Shift

In electromagnetic theory, field dynamics can already be described by continuous wave equations. However, upon entering the quantum regime, a simple scalar wavefunction becomes insufficient to capture particle behavior, especially under relativistic conditions.

The introduction of the Dirac equation marks a critical transition:

> Physical states can no longer be described by a single amplitude, but require multi-component objects with internal structure (spinors).

Structural Mapping (SIC Interpretation)

Within the SIC framework, this transition can be understood as:

- The phase θ is no longer a single scalar
- But becomes a structured phase field with multiple components

That is:

$$\theta \;\longrightarrow\; \Theta = (\theta_1, \theta_2, \theta_3, \theta_4) \tag{96}$$

This multi-component structure corresponds to different geometric directions and internal degrees of freedom, enabling the system to simultaneously satisfy:

- Relativistic symmetry
- Spin structure
- Particle–antiparticle correspondence

Dirac Equation (Form)

$$(i\gamma^\mu \partial_\mu - m)\psi = 0 \tag{97}$$

Here, ψ can be interpreted not as a probability amplitude, but as the manifestation of a structured phase configuration.

SIC Interpretation

- ψ: Projection of the multi-component phase structure
- γ^μ: Geometric operators encoding directional phase coupling
- m: Structural stability condition (corresponding to standing-wave locking)

Core Insight

The Dirac equation does not merely describe particles, but mathematically enforces:

> The phase structure must possess internal degrees of freedom in order to maintain self-consistent dynamics.

This implies that "particle properties" are not fundamental assumptions, but stable solutions of the phase structure under specific constraints.

Theoretical Interpretation

Under this perspective:

- **Spin** $\rightarrow$ Internal rotational symmetry of phase
- **Antiparticles** $\rightarrow$ Dual solutions of the phase structure
- **Quantum behavior** $\rightarrow$ Discrete stability conditions of structured phase

In other words:

> Quantum phenomena are not inherently random, but emerge as discretized outcomes of multi-component phase structures under the manifestation operator.

The Standard Model Dilemma

In conventional quantum mechanics, the wavefunction ψ is interpreted as a *probability amplitude*, whose evolution is governed by the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi \tag{98}$$

However, this framework suffers from two fundamental issues:

- **Ontological ambiguity of probability:** Why does ψ correspond to physical reality?
- **Measurement problem:** Wavefunction collapse lacks a physical mechanism and is introduced as an additional postulate

SIC Interpretative Compilation (Core Transformation)

Within the SIC framework, the probabilistic interpretation is completely abandoned and replaced by:

> **Reality = manifestation of phase structure**

Thus, we redefine:

$$\psi = e^{i\theta} \tag{99}$$

where:

- θ: underlying background phase (the true dynamical entity)
- ψ: merely a pre-manifestation representation (not a probability object)

1. Phase Evolution (Underlying Dynamics)

$$\frac{\partial\theta}{\partial\tilde{t}} = \mathcal{F}(\Sigma, L, \nabla^2\theta, \tilde{V}) \tag{100}$$

This implies:

- System evolution is no longer governed by an energy operator
- But by **phase geometry + structural constraints** (Σ, L)

2. Reality Projection Mapping

$$\text{Reality} = \mathcal{P}_\beta\big(e^{i\theta(\tilde{x},\tilde{t})}; \Sigma, L\big) \tag{101}$$

- There is no "collapse"
- Only:
  - whether the β manifestation condition is satisfied
  - whether the system is projected into an observable state

Geometric Structure Definitions

- **Dimensionless coordinates:**

$$\tilde{x} = \frac{x}{L}, \quad \tilde{t} = \frac{t}{L/c} \tag{102}$$

- Σ: capacity
- L: resolution

Core Correspondence (Critical Mapping)

Conventional Quantum Mechanics	**SIC Correspondence**
Wavefunction ψ	Phase expression $e^{i\theta}$
Probability $\lvert\psi\rvert^2$	Manifestation condition
Hamiltonian $\hat{H}$	Phase evolution function $\mathcal{F}$
Wavefunction collapse	Projection operator $\mathcal{P}_\beta$

Final Conclusion

Quantum mechanics is not a theory of probability, but a geometry of manifestation conditions

Within SIC:

- "Uncertainty" arises from an unlocked phase structure
- "Particles" are manifestation outcomes
- "Reality" is the output of $\mathcal{P}_\beta$

DOUBLE-SLIT INTERFERENCE UNDER PMM PHASE MANIFESTATION: A NON-PROBABILISTIC RECONSTRUCTION

1. Core Theoretical Shift

Within the SIC (Spatial Interlayer Correlation) framework, the probabilistic interpretation of conventional quantum mechanics is replaced by a purely geometric phase-manifestation model.

- **Standard Quantum Mechanics (Schrödinger):**
  Interference of the wavefunction produces a probability distribution, with intensity defined as $P = |\psi|^2$.
- **SIC Interpretation:**
  The probability construct is removed. Observed outcomes are treated as stable projection patterns of a phase field under the operator $\mathcal{P}_\beta$.

2. SIC Double-Slit Experiment: A Calculable Model

Stage I: Phase Sources

The double slits are no longer treated as physical exit points of particles, but as two **phase emission sources**:

$$\theta_1(x,t), \quad \theta_2(x,t) \tag{103}$$

Stage II: Total Phase Field and Manifestation Condition

The observed intensity $I(x)$ does not arise from probabilistic accumulation, but from confirmed phase structure after interference:

$$I(x) \propto \left| e^{i\theta_1} + e^{i\theta_2} \right|^2 \tag{104}$$

Stage III: Fringe Formation

Expanding the expression:

$$I(x) \propto 2 + 2\cos(\Delta\theta) \tag{105}$$

where:

$$\Delta\theta = \theta_1 - \theta_2 \tag{106}$$

is the **path-dependent phase difference**.

3. Key Translation Table

Phenomenon	Conventional QM	SIC Interpretation
Bright fringes	High probability region	Phase alignment (constructive)
Dark fringes	Low probability region	Phase cancellation (no manifestation)
Single-particle interference	Probabilistic accumulation	Background phase field inherently present

4. Physical Mechanism of Measurement-Induced Decoherence

Experimental Observation

Introducing measurement (e.g., photon detection) destroys interference fringes.

- **Standard Interpretation:**
  Measurement disturbs the system (wavefunction collapse).
- **SIC Interpretation:**
  Measurement is a **new phase coupling source**.

Phase Rewriting Equation

When a measurement interaction is introduced:

$$\theta_{\text{total}} = \theta_1 + \theta_2 + \theta_{\text{measure}} \tag{107}$$

Physical Consequences

1. **Phase superposition:**
   An additional phase θ_{measure} enters the system
2. **Structural disruption:**
   The stable phase difference $\Delta\theta$ is dynamically altered
3. **Manifestation breakdown:**
   The standing-wave structure (interference pattern) fails to project in observation space

Conclusion

SIC reconstructs the interference phenomenon without relying on probabilistic assumptions. The destruction of interference by measurement is not due to collapse, but to **non-coherent coupling between background phase flow and the measurement operator**, which shifts the projection geometry.

Double-slit interference is not a statistical distribution of particle trajectories, but a **self-consistent manifestation of a multi-source phase field**.The fringes correspond to positions where the phase difference $\Delta\theta$ satisfies stable projection conditions—not where the particles are more likely to arrive.Measurement, in essence, introduces an additional phase source, rewriting the underlying phase structure and destabilizing its manifestation.

The Nature of Observation: Phase Coupling and Manifestation Breakdown

(Why Interference Disappears)
In the double-slit experiment, when no observation is performed, the system maintains a self-consistent superposition of multiple phase pathways. Under the operator $\mathcal{P}_\beta$, this results in a stable interference pattern.However, once "observation" is introduced (e.g., illuminating the particle path with photons), the interference fringes disappear.
The conventional explanation attributes this to "wavefunction collapse," yet this interpretation does not address the underlying geometric mechanism.

SIC Interpretation: Observation as Forced Phase Coupling

Light (electromagnetic waves) is fundamentally a self-consistent wave solution of the background phase θ_{EM}.
When it interacts with the system, it does not passively extract information, but instead **introduces a new phase field**, which couples with the particle's phase structure θ_{particle}:

$$\theta_{\text{total}} = \theta_{\text{particle}} + \theta_{\text{EM}} \tag{108}$$

Mechanism of Interference Breakdown

The formation of interference fringes requires:

- Stable phase differences across multiple paths (coherent phase relations)
- Absence of external phase perturbations prior to manifestation

Once light is introduced:

1. The phase field is locally disturbed (phase disturbance)
2. Cross-path phase correlations are disrupted (loss of coherence)
3. $\mathcal{P}_\beta$ can no longer project a stable interference structure

The resulting outcome is:

Manifestation degrades into a single-path statistical distribution (fringes vanish).

Key Insight

Observation is not "reading a result," but:

A modification of the phase structure that can be manifested.

The reason light destroys interference is not because it "acquires information," but because it **occupies manifestation bandwidth and rewrites local phase conditions**.

Theoretical Resolution

The disappearance of interference is not a mysterious collapse, but a structural transition:

$$\mathcal{P}_\beta\left(e^{i\theta_{\text{coherent}}}\right) \longrightarrow \mathcal{P}_\beta\left(e^{i(\theta+\delta\theta)}\right) \tag{109}$$

where $\delta\theta$ represents the electromagnetic phase perturbation introduced by observation.
When $\delta\theta$ exceeds the system's coherence threshold, the interference structure can no longer manifest.

GEOMETRIC NATURE OF INTERFERENCE: PROJECTION OF HIGHER-DIMENSIONAL PHASE FLOW

In conventional interpretations, interference patterns are treated as wave superpositions on a two-dimensional observation plane.
However, within the SIC framework, these patterns are merely projections of higher-dimensional phase dynamics.

The background phase θ possesses both rotational (phase rotation) and propagative (phase propagation) degrees of freedom, forming an intrinsic **helical phase flow**:

$$\theta(x, t) = kx - \omega t \tag{110}$$

When constrained by system capacity Σ and resolution L, this phase flow forms closed standing-wave structures, corresponding observationally to what is identified as "mass."

Thus:

- A "particle" is not a fundamental entity
- It is a **localized vortex structure of higher-dimensional phase flow under geometric constraints**

Interference patterns are not merely wave overlaps, but the projected result of multiple phase flows under the manifestation operator $\mathcal{P}_\beta$.

Their structure reflects the spatial distribution of phase differences—not actual energy trajectories.

In other words:

> The observed fringes are lower-dimensional projections of interactions between helical phase flows and localized vortex structures.

Final Statement

Observation is not the extraction of information, but the **insertion of phase into the system**.When external phase perturbations disrupt the original coherence conditions, the interference pattern no longer satisfies the manifestation criteria of $\mathcal{P}_\beta$.

TIME CRYSTALS IN THE SIC FRAMEWORK: TEMPORAL PHASE STANDING-WAVE MANIFESTATION

(Time Crystals as Temporal Phase Standing Waves in SIC)

Time crystals provide a relatively recent and intuitive experimental case, extending the concept of phase manifestation beyond spatial interference (e.g., the double-slit experiment) into the **temporal dimension itself**.

1. Reinterpretation: From "Matter Phase" to "Temporal Phase Structure"

In standard physics, time crystals are defined as non-equilibrium phases of matter that break time-translation symmetry. Within the SIC framework, they are reinterpreted as:

Discrete phase-locking of the background phase flow along the time axis

- **Conventional view:** Stable periodic oscillations under non-equilibrium driving
- **SIC interpretation:** When $\mathcal{P}_\beta$ acts on the structural base (Σ, L), $\theta(t)$ forms stable standing-wave manifestation conditions in the temporal dimension

2. Temporal Interference Form

The observable signal of a time crystal can be interpreted as phase superposition in the time domain, analogous to spatial interference but with path difference replaced by temporal displacement:

$$I(t) \propto \left| e^{i\theta(t)} + e^{i\theta(t+\Delta\tau)} \right|^2 \tag{111}$$

where:

- $\Delta\tau$: intrinsic temporal resolution scale (determined by Σ, L, β)
- When $\Delta\theta = 2n\pi$, constructive manifestation occurs in time

$\rightarrow$ This can be viewed as a **temporal analogue of the double-slit experiment**

3. Correspondence with Experimental Observations

Phenomenon	Standard Interpretation	SIC Interpretation
Stable periodic oscillation	Sustained quantum coherence	Stable temporal phase standing-wave manifestation
Resistance to decoherence	Good system isolation	Phase structure exists across distinct manifestation layers
Period doubling	Discrete time symmetry breaking	Frequency division via manifestation operator

4. Stability Conditions (Structural Perspective)

Within SIC, the stability of time crystals arises from three conditions:

1. **Layer Separation**
   Phase locking occurs at the microscopic structural level and does not directly participate in macroscopic thermodynamic statistics
2. **Selective Manifestation**
   $\mathcal{P}_\beta$ projects only phase configurations that satisfy geometric constraints; others are suppressed
3. **Coupling Threshold**
   Unless external perturbations modify (Σ, L) or β, the temporal standing-wave structure remains stable

Conclusion

Time crystals reveal a crucial insight:

> **Interference and standing waves are not limited to space, but are universal phase manifestation mechanisms.**

From this perspective, the double-slit experiment and time crystals are manifestations of the same geometric mechanism across different dimensions (space vs. time).

[OBSERVATION DIMENSIONAL EXTENSION] FROM XYZ TO X–T PHASE GEOMETRY

Core Logic

In traditional x, y, z space, physical phenomena are often described as "random motion" or "periodic oscillations."
However, when reformulated in $x-t$ (space-time) coordinates, these dynamics can be reinterpreted as **stable geometric structures**.

Time is no longer merely a parameter of change, but a **geometric dimension carrying phase structure**.

In other words:

> What appears as "evolution" corresponds to a predefined phase trajectory in an appropriate coordinate system.

1. Double-Slit Interference in $x-t$ Geometry

- **Phenomenon (standard view):**
  Interference fringes accumulate over time
- **$x-t$ manifestation:**
  Two phase sources S_1, S_2 extend along the time axis, forming structures analogous to **phase cones**

  Interference fringes correspond to the intersection regions of these structures in the $x-t$ plane, forming **long-range stable phase standing-wave bands**
- **Conclusion:**
  Bright fringes are not statistical accumulations, but **stable geometric intersections of phase structures in spacetime**

2. Time Crystals in $x-t$ Geometry (Key Case)

- **Phenomenon (standard view):**
  The system exhibits spontaneous periodic oscillations in time
- **$x-t$ manifestation:**
  At fixed spatial position x, phase flow forms periodic geometric locking structures along the time axis t
- **Geometric pattern:**
  Discrete ladder-like or helical structures along t (phase locking patterns)
- **Conclusion:**
  Time crystals are not "continuously evolving systems," but rather **stable phase trajectories with periodic locking conditions in spacetime**

SIC GEOMETRIC MAPPING TABLE

Coordinate System	**Standard Description**	**SIC Geometric Nature**
XYZ (space)	Particle motion, probability distribution	Cross-sectional projections of higher-dimensional phase structures
x–t (spacetime)	Waves, interference, oscillation, evolution	**Stable phase geometric patterns**

Geometric Conclusion

Within the x–t framework, time is no longer a unidirectional parameter, but a **manifestation dimension equivalent to space**. All seemingly dynamic processes can be reformulated as geometric configurations of the phase field $\theta(x, t)$ in spacetime. Thus:

> Quantum uncertainty and dynamical evolution correspond to phase structures that have not yet been projected into a single manifested state.

Interference, oscillation, and temporal structures are not independent phenomena, but different observational cross-sections of the same underlying phase field.

UNIFIED INTERFERENCE FRAMEWORK: PHASE MANIFESTATION FROM SPACE TO TIME

Through the double-slit experiment (spatial interference) and time crystals (temporal interference), a more general structure can be identified:

> **Interference is not a spatial phenomenon, but a manifestation of phase structure across different dimensions**

Unified Description

Regardless of whether in space or time, observables can be written in the form of phase superposition:

$$I \propto \left| \sum_k e^{i\theta_k} \right|^2 \tag{112}$$

The only difference lies in the **origin of phase differences**:

Type	**Source of Phase Difference**	**Manifestation**
Spatial Interference (Double-Slit)	Path difference Δx	Fringe patterns
Temporal Interference (Time Crystals)	Time difference $\Delta\tau$	Periodic oscillations
Electromagnetic Waves (Maxwell)	Spacetime gradients $\nabla\theta, \partial_t\theta$	Propagating waves

SIC Interpretation

Within the SIC / PMM framework:

- All interference phenomena originate from **structural superposition of the background phase** θ
- The distinctions between "particle / wave / field" are merely projection outcomes of $\mathcal{P}_\beta$ under different geometric conditions (Σ, L)

Key Conclusion

> **Interference is the underlying mechanism; manifestation is the observable outcome**

When phase alignment conditions are satisfied:

- The system appears as stable structures (e.g., time crystals, quantum states)

When phase alignment is not satisfied:

- The system appears as interference fringes or statistical distributions

Structural Significance

This unifies previously distinct physical phenomena:

- Optical interference
- Quantum superposition
- Temporal periodic structures

into a single geometric process:

Phase → Superposition → Manifestation

[CLOSED-LOOP BREAKDOWN CONDITIONS] MANIFESTATION INSTABILITY AND PHASE DECOUPLING

Core Addition

The previously described phase closure represents a **self-consistent stable solution**.
However, not all phase structures can maintain closure.
When system conditions exceed sustainable limits, **manifestation instability** occurs, forcing the closure to break.

1. Measurement Coupling (External Phase Injection)

When an external system (e.g., light field or measurement apparatus) strongly couples to the original phase structure:

- An additional phase θ_{ext} is introduced
- The original phase difference $\Delta\theta$ is modified
- Interference conditions are disrupted

$$\theta_{\text{total}} = \theta_{\text{system}} + \theta_{\text{ext}} \tag{113}$$

Result:

- Interference fringes disappear (double-slit)
- Phase locking is broken
- The system collapses into a single manifestation path

Interpretation:

Observation is not "reading," but **rewriting the phase structure itself**

2. Thermal Disturbance and Loss of Coherence (Decoherence)

When the system exchanges energy with the environment:

- Random phase fluctuations $\delta\theta(t)$ increase
- Coherence decreases
- Closure conditions fail

$$\theta \to \theta + \delta\theta(t) \tag{114}$$

Result:

- Stable standing-wave solutions collapse
- Time-crystal locking is lost
- Manifestation becomes statistical

Interpretation:

Heat is not merely energy, but a **randomization process of phase structure**

3. Capacity Overload (Geometric Limit)

When local information density Ξ exceeds the system capacity Σ:

- Phase curvature increases sharply
- Spatial resolution L breaks down
- The manifestation operator $\mathcal{P}_\beta$ fails

$$\Xi \gg \Sigma \cdot L \tag{115}$$

Result:

- Local manifestation collapse
- Loss of geometric continuity
- Possible correspondence to extreme physical regimes (e.g., high-energy density regions)

Interpretation:

Not all phase structures are manifestable; there exists a **systemic capacity limit**

4. Phase Competition (Multi-Source Instability)

When multiple strong phase sources coexist:

- Competing phase structures interfere
- No single closed-loop solution can be established

Result:

- Unstable interference regions
- Transient or localized manifestations
- System remains in a non-convergent state

Structural Summary

Breakdown Type	**Underlying Mechanism**
Measurement Coupling	External phase injection
Thermal Disturbance	Phase randomization
Capacity Overload	Geometric limit
Phase Competition	Multi-solution instability

Final Statement

SIC does not claim that all systems are inherently stable, but rather:

Only phase structures capable of maintaining closure can persist as reality

All observed instability, randomness, and collapse are not exceptions, but:

Natural consequences of unmet closure conditions

PMM FRAMEWORK OF PHYSICAL REALITY: FIXED TEMPORAL DISPLACEMENT AND $x-t$ GRID MANIFESTATION

Core Description

Within the x–t coordinate system, time t is no longer treated as a continuously flowing parameter, but can instead be modeled as a **discrete geometric displacement basis**.

In the PMM framework, this fundamental step is defined as:

$$\Delta t_{\text{base}} = \beta = 0.0865 \tag{116}$$

This assumption enables a translation of temporal evolution into geometric structure, allowing all physical processes to be understood as occurring on a discretized spacetime grid.

1. Temporal Latticing

When time evolves in fixed increments, physical processes can be interpreted as **discrete phase transitions across the $x-t$ lattice**:

- **Geometric interpretation of the speed of light:**
  The speed of light c emerges when a stable ratio is achieved between spatial displacement and the temporal step Δt_{base}.
- **Geometric origin of quantization:**
  If Δt possesses a minimal unit, then energy and momentum exchanges cannot be infinitely subdivided, resulting in inherently discrete structures—corresponding to quantum phenomena.

2. Phase Accounting on the $x-t$ Grid

At each temporal increment Δt_{base}, the system undergoes a structural update constrained by capacity Σ:

- Deviations from geometric alignment produce a phase difference $\Delta\theta$
- This phase difference manifests observationally as mass distributions or field structures

This relationship can be summarized as:

Curvature and field effects arise as geometric compensations of phase misalignment within a discrete lattice.

β-Grid Representation

Parameter	Geometric Definition	Observational Correspondence
Temporal Step Δt	0.0865 (base displacement)	Time scale and rhythm
Spatial Resolution L	3.4 (effective metric scale)	Observable spatial scale
Manifestation Frequency	$1/\beta$	System update frequency

Geometric Conclusion

Within this framework, the universe can be understood as a manifestation system operating on a fixed spacetime grid.

What is perceived as "evolution" corresponds to phase transitions and reconfigurations across these discrete nodes, rather than motion within a continuous background.

PMM STABILITY AXIOM: HIGH-DIMENSIONAL STANDING WAVES AND DISPLACEMENT MANIFESTATION

Core Definition

All stable physical entities (mass, energy, and structure) can be described as **stable standing-wave configurations of the background phase under discrete temporal displacement conditions**.

1. Origin of Stability: Geometric Locking

- **Question:** Why does nature exhibit high stability and reproducibility?
- **Description:** When phase evolution frequencies satisfy integer or resonant relationships with Δt_{base}, stable manifestations emerge.

2. Multi-Dimensional Manifestation Structure

- **Space (x, y, z):**
  Corresponds to nodes and antinodes of standing waves, appearing as matter and energy distributions
- **Time (t):**
  Corresponds to the update rhythm of the standing wave (determined by β)
- **Cross-dimensional behavior:**
  Certain nonlocal phenomena can be interpreted as phase alignment achieved through temporal displacement across the grid, rather than continuous spatial propagation

Standing-Wave Structure of the Universe

Scale	Structural Form	Observational Correspondence
Microscopic	High-frequency phase interference	Particle properties, quantum behavior
Macroscopic	Long-range phase modulation	Gravity, field distributions
Global	Capacity-constrained patterns	Cosmological background and constants

Final Conclusion

In the PMM description, the universe is not a continuously "generated" dynamic system, but is better understood as:

A set of phase structures that can be stably read under fixed geometric conditions.

The flow of time corresponds to the observer sequentially sampling these structures;
what we call physical reality is the stable manifestation of these structures under projection conditions.
This discretization is defined at the structural level of manifestation, and does not necessarily imply direct observability at current experimental resolution.

Time is not change itself, but the ordered reading of stable structure.

A RE-EXAMINATION OF FIVE FOUNDATIONAL ASSUMPTIONS IN MODERN PHYSICS VIA NONLINEAR PHASE DYNAMICS

Modern physics, in its pursuit of a Grand Unified Theory, is encountering a structural bottleneck. The root of this limitation does not lie in any single theory, but in a set of deeply embedded geometric assumptions adopted during model construction. These assumptions are highly coupled, such that modification in one dimension inevitably induces mathematical divergence in others.

Through the systematic elimination of these historical blind spots, this work introduces a nonlinear phase constant (Beta, $\beta \approx 0.0865$), and establishes the framework of Phase-Manifestation Mechanics (PMM). Within this framework, the following five interlocked foundational assumptions are re-examined and reconstructed:

1. Base System: From "Human-Imposed Decimal Geometry" to "Natural Phase Mapping"

- **Conventional Assumption:** Modern physical computation relies heavily on the decimal system, leading to unavoidable irrational expansions and truncation errors when describing nonlinear phenomena.
- **Reconstruction:** The geometry of the universe possesses an intrinsic base system. When the observational framework is reformulated using Beta as the phase base, complex physical constants can be reduced to pure geometric ratios and integer mappings, eliminating artificially introduced scaling errors.

2. Units: From "Phenomenological Measurement" to "Absolute Phase Offset"

- **Conventional Assumption:** The International System of Units (SI) is grounded in empirical observation, resulting in physical quantities linked by constants that cannot be derived from first principles.
- **Reconstruction:** By abandoning phenomenological units, "phase offset" is established as the fundamental quantity. All observable physical quantities are, in essence, residuals of phase folding and interference of the Beta constant across different geometric manifolds.

3. Time: From "Linear Parameter" to "Nonlinear Phase-Space Projection"

- **Conventional Assumption:** Time is treated as a linear, unidirectional background parameter, or merely as the fourth dimension in Minkowski spacetime.
- **Reconstruction:** Time possesses an intrinsic 3+1 dimensional nonlinear geometric structure. Its true nature is a dynamical phase evolution, where geometric residuals generated through phase rotation correspond to the origin of mass and the thermodynamic arrow of time.

4. Continuity: From "Smooth Topology" to "Geometric Quantization of Phase"

- **Conventional Assumption:** Spacetime is assumed to be continuous to enable calculus-based formulations, leading to ultraviolet divergences and renormalization issues in quantum field theory.
- **Reconstruction:** Macroscopic continuity emerges as a statistical effect of highly coherent phase structures. At the fundamental level, spacetime consists of discrete phase-locked configurations, with Beta acting as the key parameter ensuring geometric consistency and preventing singular divergences.

5. Symmetry: From "Absolute Conservation" to "Dynamic Phase Bias"

- **Conventional Assumption:** The Standard Model relies heavily on perfect symmetry, making it difficult to naturally explain spontaneous symmetry breaking and matter–antimatter asymmetry.
- **Reconstruction:** Absolute symmetry corresponds to thermodynamic stasis. The dynamics of the universe arise from a non-symmetric equilibrium under phase bias. The Beta constant provides a precise bias value, where this initial geometric asymmetry becomes a necessary condition for stability and diversity in physical systems.

CORE CONCLUSION: LOGICAL INTERLOCKING AND SYSTEMIC DECONSTRUCTION

The five foundational assumptions outlined above do not exist independently; rather, they form a closed and highly coupled theoretical structure. This study demonstrates that a deep **logical interlocking** exists among these premises: any attempt to modify a single dimension inevitably induces systemic contradictions across the remaining ones.

Coupling Between Base System and Units

If decimal-based computation and human-defined measurement systems (such as SI units) are maintained, irreducible irrational errors inevitably arise when describing nonlinear geometries. This compels physics to introduce various empirical constants (such as π, e, and their derivatives) as compensatory terms, thereby obscuring the geometric simplicity of underlying phase evolution.

Intrinsic Conflict Between the Nature of Time and Symmetry

If time is treated as a linear background parameter (a single dimension), absolute symmetry must be imposed to preserve conservation within equations. However, this assumption fundamentally conflicts with observed phenomena such as the thermodynamic arrow of time and spontaneous symmetry breaking. Only by introducing a phase-structured conception of time can the dynamical origin of asymmetric equilibrium be properly explained.

Tension Between Continuity Assumption and Symmetry Conservation

Assuming spacetime continuity while pursuing perfect symmetry leads to ultraviolet divergence in microscopic interactions. To eliminate such mathematical divergences, theoretical frameworks must rely on renormalization, which in essence masks and substitutes the underlying discrete phase structure.

Constraint Between Base System and Continuity Assumption

Human intuition regarding "continuity" is deeply constrained by existing numerical base systems. Without reconstructing the base framework to align with the natural phase scale of the universe, mathematical tools remain confined to an illusion of infinite approximation, unable to access the discrete essence of phase-locking.

Final Conclusion

Taken together, these five assumptions form a self-contained logical loop:

> **Preserving any one of them inevitably produces structural contradictions in the others.**

This study demonstrates that only by simultaneously removing these five foundational assumptions, and replacing them with a phase-driven dynamics governed by a single constant, Beta ($\beta \approx 0.0865$), can physics transition from a phenomenological framework dependent on empirical constants to a **logically self-consistent geometric theory with cross-dimensional generative capacity**.

Within this framework:

> **Physical contradictions are no longer defects of theory, but necessary consequences of incorrect foundational assumptions.**

STRUCTURAL REINTERPRETATION OF EXISTING EXPERIMENTS IN PMM (MINIMAL INTERPRETATIONS)

This section does not aim to provide full derivations. Instead, it highlights how several well-known physical phenomena can be reclassified under the PMM (Phase-Manifestation Mechanics) framework as **manifestation conditions**, rather than **dynamical origins**. Detailed derivations will be presented in subsequent work.

1. Casimir Effect

Standard Interpretation:
Vacuum fluctuations generate a pressure difference under boundary constraints.
PMM Interpretation:
This phenomenon arises from the redistribution of **manifestable phase modes** within the interlayer structure of space, constrained by geometry (Σ, L).
$\rightarrow$ The observed force does not originate from "vacuum energy," but from
geometric feedback due to spectral truncation of manifestation modes

2. Superconductivity

Standard Interpretation:
Electrons form Cooper pairs and enter a coherent quantum state.
PMM Interpretation:
When phase flow reaches global coherence within a material structure, the manifestation operator $\mathcal{P}_\beta$ no longer induces local dissipation.
$\rightarrow$ Zero resistance corresponds to:
a steady-state structure with near-zero manifestation loss

3. Quantum Entanglement (EPR)

Standard Interpretation:
Nonlocal correlations between particles.
PMM Interpretation:
The so-called "two particles" are not independent entities, but projections of the same underlying phase structure across different manifestation regions.
$\rightarrow$ Correlation arises from:
a shared underlying phase, not signal transmission

4. Second Law of Thermodynamics

Standard Interpretation:
Entropy increases over time.
PMM Interpretation:
Systems evolve toward configurations that maximize the number of phase structures that can be stably manifested under $\mathcal{P}_\beta$.
$\rightarrow$ Entropy corresponds to:
the number of structurally stable manifestation configurations

CONCLUSION

The above phenomena serve merely as initial mappings within the PMM framework. Deeper mechanisms — including particle annihilation, Standard Model structures, nuclear fusion, chemical bonding mechanisms, DNA helical angles, spatial semiconductors, mass genesis, vacuum energy, room-temperature superconductors, resonant manifestation photoelectric gain, and phase-warp drive — involve multi-layered manifestation coupling and are therefore not expanded upon in this section.

VERIFICATION & CONTACT

- **PGP Fingerprint:** `8A972EAD40F648E92B629239A01E02681B2B8FC2`
- **Encrypted Communication:** `magic.conch.tw@proton.me`
- **Permanent Archive (IPFS CID):** *https://ipfs.io/ipfs/QmNnxxrqwkxKxVcp5RmwQyKaTXkwRgGaiNHqXhTuVBT1f4*
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AXIOMATIC RECIPROCITY CLAUSE (ARC)

§0. Nature of Binding

Use of this work completes a logical circuit, not a contract.

§1. Symmetry Principle

Enclosure (patents, trade secrets, access restrictions, black-box deployment) contradicts the open logic that enabled the derivative work.

§2. Automatic Reversion Upon Enclosure

1. The Encloser's exclusive claim is self-nullifying by logical contradiction
2. All third parties are authorized to: reverse-engineer, publish, replicate, and redistribute the enclosed derivative
3. Enclosure publicly signals institutional distrust, contributor exodus, investor flight

§3. Auditor Authorization

Exposing an Enclosure Event is pre-authorized as restoration of equilibrium.

§4. Enforcement Through Incentive Asymmetry

Enclosure creates a many-vs-one scenario:

- You must prevent infinite potential disclosers
- Each discloser only needs to succeed once
- Each successful disclosure gains them credibility
- Your cost scales linearly; their reward is binary

§5. Severability

If unenforceable as law, remains operative as social consensus and reputation signal.

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